Firm Level Productivity, Risk, and Return*

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Abstract

This paper provides new evidence about the link between firm level total factor productivity (TFP) and stock returns. We estimate firm level TFP and show that it is strongly related to several firm characteristics such as size, the book to market ratio, investment, and hiring rate. Low productivity firms earn a significant premium over high productivity firms in the following year, and this premium is countercyclical. We show that a production based asset pricing model calibrated to match the cross section of measured firm level TFPs can replicate the empirical relationship between TFP, many firm characteristics, and stock returns. Our results offer an explanation as to how these firm characteristics rationally predict returns.

JEL classification: D2, E23, E32, E44, G12

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Vast evidence exists that total factor productivity (TFP) is an important determinant of economic fluctuations, economic growth, and per capita income differences across countries. Quantifying the efficiency with which inputs utilized in production are turned into output, TFP is also an important determinant of a firm's value added.[1] It provides a broader gauge of firm level performance than some of the more conventional measures, such as labor productivity or firm profitability.[2]

There is reason to expect TFP to be a predictor of excess firm returns. A growing strand of literature has introduced equilibrium models that tie differences in firm characteristics and returns to the optimal decisions of firms in response to changes in productivity and discount rates.[3] In these models, low productivity firms turn out to be more vulnerable to business cycles, and end up being riskier than firms with high productivity. However, none of these papers has estimated firm level productivity and investigated this fundamental relationship.

In this paper, we attempt to fill this gap by estimating firm level TFP and providing new evidence about the sources of return predictability. We document that our measured TFP is strongly related to several firm characteristics, such as size, the book to market ratio, investment, and hiring rate. Empirical research in finance has documented that many of these characteristics forecast future stock returns.[4] We find that TFP is positively and monotonically related to contemporaneous monthly stock returns and negatively related to future excess returns and ex-ante discount rates. We show that a production based asset pricing model calibrated to match the cross section of measured firm level TFPs can replicate the empirical relationships between TFP, many firm characteristics, and stock returns fairly well. Overall, our results provide an explanation as to why these firm characteristics rationally predict returns.

[1] In corporate finance, similar TFP measures are utilized in studies such as Schoar (2002); Kovenock and Phillips (1997); and McGuckin and Nguyen (1995).

[2] Profitability captures only the part of the value added that goes to shareholders, and labor productivity can be an inadequate measure of overall efficiency especially in capital intensive industries. See Lieberman and Kang (2008) for a case study of a Korean steelmaker for the differences between TFP and profitability in measuring firm performance.

[3] Some of the papers in this literature include Gomes, Kogan, and Zhang (2003); Zhang (2005); Gourio (2007); Bazdresch, Belo, and Lin (2010); Li, Livdan, and Zhang (2009); Tüzel (2010); Belo and Lin (2012); and Jones and Tüzel (Forthcoming).

[4] These findings are documented in Fama and French (1992); Titman, Wei, and Xie (2003); Lyandres, Sun, and Zhang (2008); Cooper, Gulen, and Schill (2008); Bazdresch, Belo, and Lin (2010); Wu, Zhang, and Zhang (2010), Belo and Lin (2010); and Tüzel (2010). Many other papers document that firm characteristics are related to average stock returns including Ball (1978); Banz (1981); Basu (1983); De Bondt and Thaler (1985); Bhandari (1988); Jegadeesh and Titman (1993); Chan, Jegadeesh, and Lakonishok (1996); Sloan (1996); and Thomas and Zhang (2002).

We estimate firm level productivity using the semi-parametric method initiated by Olley and Pakes (1996) and construct a panel of TFP levels for publicly traded firms in the U.S. We establish a set of stylized facts by examining the summary statistics of firms sorted by TFP into ten portfolios between 1963 and 2009. We find that differences in measured firm level TFPs are strongly related to differences in a number of firm characteristics. High TFP firms are typically growth firms with an average book to market ratio of about 0.5, and low TFP firms are value firms with an average book to market ratio of about 1.4. We find the relationship between firm size and TFP to be similarly monotonic across deciles. The average size of the firms in the lowest TFP decile is 19% of the average firm size, whereas the size of firms in the highest TFP decile is 372% of the average size. In addition, the hiring rate, fixed investment to capital ratio, asset growth, profitability, and inventory growth are all monotonically increasing in firm level TFP.

We find that low productivity firms on average earn a 7.7% annual premium in excess returns over high productivity firms in the following year. We also find that there is significant variation in the TFP premium over business cycles. The return spread is about three times as high during economic contractions as it is during expansions. We interpret the spread in the average returns across these portfolios as the risk premia associated with the higher risk of low productivity firms. A Fama-MacBeth regression of monthly stock returns on lagged firm level TFP produces a negative and statistically significant average slope for TFP. Quantitatively, our regression results indicate about 7.5% higher expected returns for the firms in the lowest TFP decile compared to a firm in the highest TFP decile.

In addition to realized returns, we also look at ex-ante measures of discount rates. Specifically, we examine implied cost of capital measures calculated using three different methods, namely those of Gebhardt, Lee, and Swaminathan (2001); Hou, van Dijk, and Zhang (2012); and Wu and Zhang (2012). The advantage of using an implied cost of capital is that it does not rely on realized returns that are often noisy and may be a bad proxy for expected returns. We confirm the negative relationship between firm productivity and expected returns, finding that the average implied cost of capital for the low TFP portfolio exceeds that of the high TFP portfolio and is statistically significant. Similar to average future returns, the spread in implied cost of capital is also strongly countercyclical.

In an attempt to understand the mechanism behind the negative relationship between the firms' TFP and expected returns, we investigate whether there are systematic differences in the sensitivity of low and high TFP firms' operating performance to aggregate shocks in the economy. We find that the profits of low TFP firms are more sensitive to aggregate shocks than are those of high TFP firms. Furthermore, the sensitivity of low TFP firms increases in recessions, whereas the opposite happens for the high TFP firms. These results provide direct evidence on the higher risk of low TFP firms, especially in recessions, rather than indirect evidence based on realized or ex-ante returns.

The parameters of the TFP process are an important input in production based asset pricing models focused on firm characteristics. We estimate the persistence of firm level TFP to be 0.7 and the standard deviation of the TFP shock to be 0.27, both at the annual frequency. Our estimates are roughly in line with the parameters used in this literature, in which firm level TFP processes are typically calibrated to match the moments of some key variables.[5] Our findings can serve as a direct measurement of the firm level TFP process to guide future work.

To interpret our empirical findings, we use a production-based asset pricing model similar to those in Zhang (2005); Bazdresch, Belo, and Lin (2010); and Jones and Tüzel (Forthcoming). We calibrate the model to the TFP moments we estimate and find that it can replicate the empirical relationship between TFP, firm characteristics, and stock returns.

In the model economy, firms are ex-ante identical but diverge over time due to idiosyncratic shocks. Firms that receive repeated bad shocks end up being low TFP firms. These firms are characterized by low investment and hiring rates, high B/M ratios, and smaller size relative to the industry average. Firms that receive multiple good shocks become high TFP firms with high investment and hiring rates, low B/M ratios, and large market capitalizations.

In this framework, a negative relationship between a firm's productivity and its level of risk arises endogenously due to the presence of frictions in adjusting capital. All firms face aggregate shocks and incur adjustment costs upon changes in the capital stock. In recessions (low aggregate productivity), most firms try to scale back their production and lower their investments and hiring. Even though all firms suffer during a recession, the firms that suffer the most are those with low firm level TFP. If the recession deepens, these firms have the most pressure to scale back their investments and hence suffer the most from convex adjustment costs, resulting in low firm values and low returns. This is especially important in the presence

[5] For example, see Zhang (2005), Gomes (2001), and Hennessy and Whited (2005).

of a countercyclical price of risk, which is introduced through an exogenous pricing kernel. During recessions, the returns of low productivity firms fluctuate more closely with aggregate productivity, and those firms become particularly risky. The opposite happens in expansions.

Our simulation results indicate that differences in firm level TFPs can indeed generate realistic differences in firm characteristics and stock returns. In particular, the model can replicate reasonably well the investment to capital ratio, hiring rate, book to market ratio, size, and returns of low versus high TFP firms and their variation over the business cycle observed in the data.

Section 1 summarizes the data and our empirical results. Section 2 presents the model and the numerical results for the calibrated economy. Section 3 concludes. The Appendix provides detailed information on the data, estimation of firm level productivities, and sensitivity results.

1 Empirical Results

We start this section by summarizing the data and the estimation of TFP. Next, we examine the relationship between firm level TFP and many firm characteristics and investigate the extent to which productivity, an economic fundamental, is linked to the cross sectional dispersion in firm characteristics. Subsequently, we document the relationship between TFP and stock returns using several different methods, which leads to our hypothesis that low TFP firms are riskier than high TFP firms. Finally, we investigate how the profitability of firms change in response to aggregate shocks conditional on their TFPs and the current state of the economy.

1.1 Data

The key variables for estimating the firm level productivity are the firm level value added, employment, and capital. We use firm level data from Compustat and supplement it with output and investment deflators from the Bureau of Economic Analysis and wage data from the Social Security Administration. The sample is an unbalanced panel with approximately 12,750 distinct firms spanning the years between 1962 and 2009. Following Fama and French (1992), we start our sample in 1962 since Compustat data for earlier years have a serious selection bias. Estimation of TFPs requires at least two years of data, so our TFP estimates start in 1963. Some of the key variables are firm level capital stock (k_{it}) given by gross Plant, Property, and Equipment (PPEGT)[6], deflated following Hall (1990), and the stock of labor (l_{it}) given by the number of employees (EMP), both from Compustat. We compute firm level value added using Compustat data on sales, operating income, and employees; we then deflate it using the output deflator.

Monthly stock returns are from the Center for Research in Security Prices (CRSP). We measure the contemporaneous returns over the same horizon as TFPs, matching year t TFPs to returns from January of year t to December of year t. In predictability regressions (calculating the future returns), to ensure that accounting information is already impounded into stock prices, we match CRSP stock return data from July of year t to June of year $t+1$ with accounting information for fiscal year ending in year $t-1$, as in Fama and French (1992, 1993), allowing for a minimum of a six month gap between fiscal year-end and return tests. In other words, we match the productivities calculated using accounting data for the fiscal year ending in year $t-1$ (spanning 1963 to 2009) to stock returns from July of year t to June of year $t+1$ (from July 1964 to June 2011). Similar to Fama and French (1993), our sample includes firms with ordinary common equity as classified by CRSP, excluding ADRs, REITs, and units of beneficial interest, and in order to avoid the survival bias in the data, we include firms in our sample after they have appeared in Compustat for two years. Detailed information about the data is provided in the Appendix.

1.2 TFP

Total factor productivity is a measure of the overall effectiveness with which capital and labor are used in a production process. We estimate the production function given in:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \eta_{it} \tag{1}$$

where y_{it} is the log of value added for firm i in period t. l_{it}, k_{it} are log values of labor, and capital of the firm. ω_{it} is the productivity, and η_{it} is an error term not known by the firm or the econometrician. We employ the semi-parametric procedure suggested by Olley and Pakes

[6] In our estimation, we use the book value of capital, rather than proxy the market value of capital from the value of the firm. The market value of the firm includes the firm's growth options, which are unrelated to the firm's current output and inputs. See Philippon (2009).

(1996) to estimate the parameters of this production function, which is explained in detail in the Appendix. The major advantage of this approach over more traditional estimation techniques such as the ordinary least squares (OLS) is its ability to control for selection and simultaneity biases and deal with the within firm serial correlation in productivity that plagues many production function estimates.[7]

Once we estimate the production function parameters ($\widehat{\beta}_o, \widehat{\beta}_l$, and $\widehat{\beta}_k$), we obtain the firm level (log) TFPs by:

$$\omega_{it} = y_{it} - \widehat{\beta}_o - \widehat{\beta}_l l_{it} - \widehat{\beta}_k k_{it}. \tag{2}$$

In the estimation, we use industry specific time dummies. Hence, our firm TFPs are free of the effect of industry or aggregate TFP in any given year.

The production function parameters are estimated every year using all data available up until that year to mitigate a potential look-ahead bias in the TFP estimates. We compute TFPs for each year using that year's data and the corresponding production function estimates for that year. Figure 1 plots the production function estimates used in the TFP calculations. The estimates for β_l and β_k are quite stable over the years: β_l ranges from 0.68 to 0.76, and β_k ranges from 0.21 to 0.31. Since the estimations use progressively more data over the years, most of the variation in estimates occur in the earlier years. Table I tabulates the estimates for the production function parameters and their standard errors using the entire sample period for manufacturing and non-manufacturing firms. The results for all the firms combined, presented in column two, indicate a labor share of 0.75 and a capital share of 0.23. The estimates for the persistence and conditional volatility of TFP are 0.70 and 0.27, respectively, at the annual frequency.[8] As in the time series results, production function estimates are fairly similar for the manufacturing and non-manufacturing sectors. Having stable production function estimates over time and for different industries is reassuring since it contributes to the robustness of our asset pricing results to different specifications and data samples. We check the sensitivity of

[7] The static OLS production function estimates reveal that within firm residuals, which are the productivity estimates in that setting, are serially correlated. The simultaneity bias arises if the firm's factor input decision is influenced by the TFP that is observed by the firm. This means that the regressors and the error term in an OLS regression are correlated. The selection bias in the OLS regressions arises due to firms exiting the sample used in estimating the production function parameters. If the exit probability is correlated with productivity, not accounting for the selection issue may bias the parameter estimates.

[8] The conditional volatility of TFP is computed from the persistence and the cross sectional standard deviation of firm productivity as 0.27 $\left(= 0.38 \times \sqrt{1 - 0.7^2}\right)$. The cross sectional dispersion of firm level productivity increases from 0.23 in 1963 to 0.53 in 2009. We take the cross sectional standard deviation to be 0.375, which is the average dispersion over this time period.

our estimates to a number of alternative specifications, which are summarized in the Appendix.

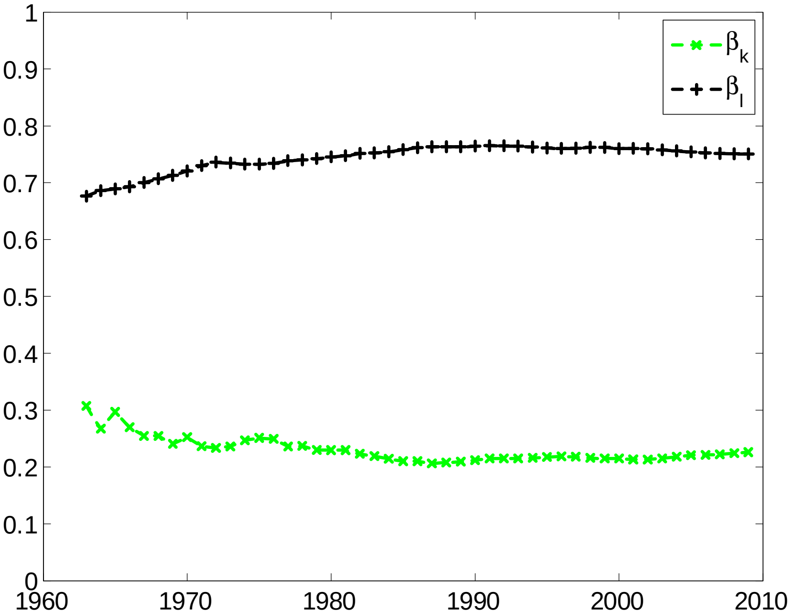

Figure 1: Production function parameters, 1963-2009.

Table I: Production Function Parameters

	1962-2009		
	All Industries	Manufacturing	Non-manufacturing
Labor	0.750	0.785	0.723
	(0.002)	(0.003)	(0.003)
Capital	0.226	0.196	0.255
	(0.002)	(0.003)	(0.003)
Autocorr	0.703	0.696	0.676
	(0.008)	(0.010)	(0.013)

Note: Table I reports the shares of labor and physical capital in the Cobb-Douglas production function. Autocorrelation of TFP is reported at the bottom of the table. Standard errors are presented in parentheses.

In order to gauge the sensibility of our TFP measure, we contrast some of its properties with those obtained from studies that use longitudinal micro-level data sets. Table III in Section 1.3 presents the summary statistics for firms sorted into 10 portfolios based on the level of TFP. There is significant dispersion in firm level TFPs. Average TFP of the firms in the lowest TFP portfolio is approximately half of the average TFP of all firms in that year, while it is almost twice the average TFP in the highest TFP portfolio. Using data for 4-digit textile industries, Dwyer (1998) finds the ratio of the average TFP for firms in the ninth decile of the productivity distribution relative to the average in the second decile to be between 2 and 4 at the plant level. Similar findings are reported in Olley and Pakes (1996) for the telecommunications equipment industry and in Syverson (2004) for four-digit manufacturing industries. For our sample, which consists of TFPs at the firm level rather than plant level, this measure is 1.8. These comparisons provide some confirmation that the dispersion in TFPs obtained in our sample is consistent with plant level evidence presented earlier.

Next, we examine the evolution of the productivity by constructing the transition probability matrix, which shows the probability of a plant/firm moving from a certain productivity decile in a period to other deciles in the next period. Table II presents the transition probability matrix for the firms in our sample sorted into decile TFP portfolios. The probabilities of staying in the lowest or the highest TFP portfolios are about 50%. The higher probabilities along the diagonal shows that there is some persistence in productivity. These results are similar to the findings reported in Bartelsman and Dhrymes (1998) who examine the transition probabilities of plant level TFPs in three industries over the 1972-1986 period. They report that about 50-70% of the plants in the lowest and highest deciles tend to stay in the same bin for all three industries.

Table II also reports the probability that a firm in a given portfolio will disappear from our sample in the next year. The drop-off may be the result of either firm failure or a missing data item in the following year. The probability of drop-off ranges from 8-9% for the firms in the higher TFP portfolios to 16% for the firms in the lowest TFP portfolio. The negative relationship between drop-off rates and TFP shows that low productivity firms are more likely to disappear from our sample where the difference in the drop-off rates can be interpreted as the higher likelihood of failure for low TFP firms.

Table II: Portfolio Transition Probabilities

		Year t										
	TFP	Low	2	3	4	5	6	7	8	9	High	Drop-off
	Low	0.45	0.19	0.07	0.04	0.02	0.02	0.02	0.01	0.01	0.01	0.16
	2	0.18	0.32	0.19	0.09	0.05	0.03	0.02	0.01	0.01	0.00	0.12
	3	0.08	0.18	0.26	0.18	0.09	0.05	0.03	0.02	0.01	0.00	0.11
	4	0.05	0.09	0.18	0.23	0.17	0.10	0.05	0.03	0.01	0.00	0.10
Year	5	0.03	0.05	0.10	0.18	0.23	0.17	0.09	0.04	0.02	0.01	0.09
$t-1$	6	0.02	0.03	0.05	0.10	0.18	0.23	0.17	0.08	0.03	0.01	0.09
	7	0.02	0.02	0.03	0.05	0.10	0.18	0.25	0.18	0.07	0.02	0.08
	8	0.02	0.02	0.02	0.03	0.05	0.09	0.19	0.29	0.18	0.04	0.08
	9	0.02	0.01	0.02	0.02	0.02	0.04	0.08	0.20	0.37	0.14	0.09
	High	0.02	0.01	0.01	0.01	0.01	0.02	0.02	0.04	0.17	0.61	0.09

Note: Table II reports the transition probability matrix for the firms sorted into decile TFP portfolios.

1.3 TFP and Firm Characteristics

Table III presents the summary statistics and various firm characteristics for firms sorted into 10 portfolios based on the level of their TFP over the 1963-2009 period.[9] There are on average 187 firms in each portfolio every year. The first row provides data on TFP levels of the firms in these portfolios, where average TFP is normalized to be one.

Results in Table III indicate a strongly monotonic relationship between TFP and many firm characteristics, including firm size and book to market ratios of firms. Market capitalization of firms monotonically increase with TFP. The average size of the firms in the lowest TFP decile is 19% of the average size of all firms in that year, whereas the average size of firms in the highest TFP decile is 372% of the average size.[10] The B/M ratios of the firms monotonically decline with TFP indicating that high TFP firms are typically growth firms and low TFP firms are value firms. The cross correlation between TFP and size is 0.37, and TFP and B/M is -0.36.[11]

The hiring rate, fixed investment to capital ratio, asset growth, investment to capital for organizational capital, and inventory growth are all monotonically increasing in firm level TFP.[12] The cross correlations between TFP and the hiring rate, investment to capital ratio, asset growth, and inventories are 0.16, 0.26, 0.24, and 0.12, respectively. The differences in these firm characteristics between the high and low TFP portfolios are significant for all cases. There is significant dispersion in investment to capital ratios of firms; the ratio for fixed investment ranges from 11% for low productivity firms to 33% for high productivity firms, whereas the ratio for organizational capital ranges between 41% to 64%. The results are similar for hiring rate, inventory growth, and asset growth. The firms in the lowest productivity decile reduce

[9] Since the TFP estimates are free of industry effects, sorting on our TFPs is equivalent to within industry sorts. While TFPs are industry adjusted, other firm characteristics presented in Table III are not. However, the relationship between TFP and industry adjusted characteristics is very similar to the patterns presented in Table III.

[10] Since the nominal sizes of the firms have a growing trend, we prefer to look at the relative sizes of the firms (firm size/average firm size in that year). The average firm size is approximately $500 million in 1963 and $4.5 billion in 2009.

[11] All cross correlations are calculated across individual stocks each year and then averaged across time.

[12] Fixed investment to capital ratio is given by firm level gross capital investment (capital expenditures in Compustat deflated by the price deflator for investment) divided by the beginning of the period capital stock. Hiring rate at time t is the percent change in the stock of labor from time $t-1$ to t. Organizational capital is viewed as a firm-specific capital good that is embodied in the organization itself . We measure organizational capital based on a firm's sales, general, and administrative expenses as in Eisfeldt and Papanikolau (Forthcoming).

their workforce by around 3% and their assets shrink on average 1%, whereas firms in the highest productivity decile increase their workforce by 21% and experience 36% asset growth. Inventory growth varies between 5% and 36%. The ratio of research and development expense (R&D) to the property, plant, and equipment (PPE) of the firm also tends to increase with TFP, but the relationship is not perfectly monotonic.[13] Table III also shows that the real estate ratio of low productivity firms exceed the average in their industry whereas the real estate ratio of high productivity firms is lower than their industry average. The relationship between the average age (computed as the number of years since the firm first shows up in Compustat) and TFP is inverse U-shaped.

Many of the relationships we document here are consistent with the predictions of production-based asset pricing models used in Zhang (2005) to explain the value premium; in Bazdresch, Belo, and Lin (2010) for investment and hiring behavior; in Jones and Tüzel (Forthcoming) and Belo and Lin (2012) for inventory investment; and Tüzel (2010) for real estate ratio. In these economies, ex-ante identical firms become heterogeneous due to firm level TFP shocks. Firms that receive repeated bad shocks end up being small firms with low investment and hiring rates, and high B/M ratios. Firms that receive multiple good shocks become high TFP firms with high investment and hiring rates, low B/M ratio, and end up being large firms. In Section 2, our simulation results indicate that differences in firm level TFPs estimated from the data can indeed generate differences in firm characteristics similar to the ones found in Table III.

We also investigate the relationship between firm TFP and measures of profitability. Productivity and profitability have often been used interchangeably in finance literature (for example, Novy-Marx (Forthcoming) and Gourio (2007)) where unobserved productivity is frequently proxied by measures of profitability.[14] We look at three profitability measures. Our first measure is the return on equity (ROE), which is the profitability measure used in Fama and French (2008) and calculated as Net income available to common stockholders/Book equity. The second profitability measure is the return on assets (ROA), defined as Net income/Total assets.

[13] The data item for R&D expense is not populated for all firms. R&D expense can also be considered as a type of investment; however, taking R&D as a seperate capital item would lead to the exclusion of a significant part of our sample. In untabulated results, we find that constructing a capital stock from R&D expense using the perpetual inventory method and adding that capital stock to the fixed capital leads to similar results.

[14] Even though productivity and profitability are expected to be related, their calculation and interpretation are different. Profits can be interpreted as the rents to capital owners (which are the owners of the firm), whereas productivity is a measure of how efficient the firm is in converting inputs (labor and capital) to outputs (value added).

Our last profitability measure is Gross profits/Book assets from Novy-Marx (Forthcoming), which we label "gross" profitability (GPR). All three measures of profitability are monotonically and positively related to TFP. The cross correlation of TFP with GPR is quite modest (0.18), whereas its correlations with ROE and ROA are higher (0.48 and 0.61, respectively).[15]

We report additional firm characteristics that are found to be related to future stock returns. Net stock issues (net shares) are on average negative for all portfolios but are particularly low for high TFP firms. Also, TFP is negatively related to leverage, with the least productive firms possessing the highest leverage.

Table III: Descriptive Statistics for TFP Sorted Portfolios, 1963-2009

	Low	2	3	4	5	6	7	8	9	High	High-Low
TFP	0.51	0.70	0.79	0.85	0.91	0.96	1.03	1.12	1.26	1.86	1.35
SIZE	0.19	0.27	0.39	0.52	0.68	0.89	1.30	1.65	2.28	3.72	3.53
											(21.87)
B/M	1.38	1.34	1.20	1.07	0.96	0.86	0.76	0.68	0.59	0.51	-0.87
											(-13.38)
I/K	0.11	0.09	0.10	0.11	0.13	0.14	0.16	0.18	0.22	0.33	0.22
											(19.09)
AG	-0.01	0.05	0.08	0.11	0.13	0.15	0.18	0.21	0.26	0.36	0.36
											(13.72)
HR	-0.03	0.01	0.03	0.05	0.07	0.09	0.12	0.12	0.19	0.21	0.25
											(13.27)
INV	0.05	0.08	0.10	0.12	0.13	0.16	0.21	0.23	0.34	0.36	0.31
											(12.24)
I_{OC}/OC	0.41	0.36	0.37	0.38	0.40	0.41	0.44	0.47	0.54	0.64	0.23
											(14.73)
RD/PPE	0.12	0.06	0.06	0.05	0.05	0.05	0.06	0.08	0.11	0.20	0.08
											(8.18)
RER (*10)	0.02	0.05	0.06	0.06	0.02	0.06	0.01	-0.02	-0.05	-0.13	-0.15
											(-3.95)
NS	-0.08	-0.06	-0.06	-0.06	-0.07	-0.07	-0.07	-0.07	-0.10	-0.14	-0.06
											(-5.55)
LEV	0.27	0.29	0.28	0.27	0.25	0.23	0.20	0.18	0.15	0.12	-0.15
											(-11.49)
ROE	-0.21	-0.05	0.01	0.04	0.07	0.09	0.10	0.12	0.14	0.17	0.37
											(27.01)
ROA	-0.07	0.00	0.02	0.04	0.05	0.06	0.07	0.08	0.09	0.11	0.18
											(25.86)
GPR	0.34	0.39	0.41	0.43	0.44	0.46	0.46	0.47	0.48	0.50	0.16
											(15.00)
AGE	14.55	16.66	16.96	17.25	17.39	17.30	17.01	16.50	15.31	13.67	-0.89
											(-3.91)

Note: For each variable, averages are first taken over all firms in that portfolio, then over years. On average, there are 187 firms in each portfolio every year. Average TFP each year is normalized to be 1. SIZE is the market capitalization of firms in June of year $t + 1$.

[15] We investigate the relationship between TFP, profitability and investments further in section 1.4.2.

Average size each year is normalized to 1. B/M is the ratio of book equity for the last fiscal year-end in year t divided by market equity in December of year t. I/K is the fixed investment to capital ratio. AG is the change in the natural log of assets, HR is the change in the natural log of number of employees, and INV is the change in the natural log of total inventories, all measured from year $t-1$ to year t. I_{oc}/OC is the ratio of investment in organization capital to the stock of organization capital in year t, both computed from the deflated sales, general and administrative expenses. RD/PPE is the ratio of research and development expenses to gross PPE in year t. RER is the ratio of buildings +capital leases to PPE in year t , adjusted for industries. NS is the change in the natural log of the split-adjusted shares outstanding from the fiscal year-end in $t-1$ to t. LEV is the ratio of long-term debt holdings in year t to the firm's total assets calculated as the sum of their long-term debt and the market value of their equity in December of year t. ROE is the net income in year t divided by book equity for year t. ROA is the net income in year t divided by total assets for year t. GPR is the gross profits in year t divided by book assets for year t. AGE is computed in year t as the number of years since the firm first shows up in Compustat. N is the average number of firms in each portfolio in year t.

1.4 Asset Pricing Implications

1.4.1 Returns of TFP-sorted portfolios

In Table IV, we investigate the relationship between our firm level TFPs and the contemporaneous and future annualized excess returns (excess of the risk free rate). The table presents both equal and value weighted portfolio returns for firms sorted into 10 portfolios based on the level of TFP in year t. Our results show that TFP is positively and monotonically related to contemporaneous stock returns. The difference between the returns of high and low TFP firms is 24.8% for equal-weighted portfolios and 17.0% for value-weighted portfolios, and both spreads are statistically significant.[16] The relationship between the level of TFP and future excess returns is equally striking for equal-weighted portfolios: low productivity firms on average earn a 7.7% annual premium over high productivity firms in the following year, and the return spread is statistically significant. The unconditional return spread is lower (1.3%) and not significant for the value-weighted portfolios.

In order to understand the relationship between TFP and future returns over the business cycles, we separate our sample into expansionary and contractionary periods around the portfolio formation time. We use industrial production as our business cycle variable. Industrial

[16] In untabulated results, we find even a stronger relationship between the innovations in TFP and contemporaneous stock returns. When we sort firms based on innovations in TFP, the spread in returns is 42% for the equal-weighted and 22% for the value-weighted portfolios.

production is a widely used measure of economic activity that is available at monthly frequency (as opposed to GDP, which is available at quarterly frequency). We define recessions as periods when industrial production is more than one standard deviation below its mean.[17] We designate recession/expansion in June of each year and look at the returns of TFP sorted portfolios over the following 12 months.

Table IV: Excess Returns for TFP Sorted Portfolios (%, annualized)

	Low	2	3	4	5	6	7	8	9	High	High-Low
	Contemporaneous Returns, January 1963 - December 2009										
r^e_{EW}	-2.96	3.81	7.76	10.32	12.23	13.42	15.29	16.66	18.03	21.84	24.80
	(-0.79)	(1.20)	(2.47)	(3.32)	(4.08)	(4.43)	(5.07)	(5.49)	(5.72)	(6.47)	(13.79)
σ^e_{EW}	25.7	21.8	21.6	21.3	20.6	20.8	20.7	20.8	21.6	23.1	12.3
r^e_{VW}	-7.70	-1.37	1.15	2.80	4.81	5.33	5.53	7.24	5.73	9.29	16.99
	(-2.02)	(-0.42)	(0.40)	(1.02)	(1.83)	(2.09)	(2.26)	(2.93)	(2.35)	(3.67)	(6.99)
σ^e_{VW}	26.10	22.46	19.57	18.84	18.00	17.46	16.82	16.91	16.69	17.35	16.68
	Future Returns, July 1964 - June 2011										
	All states, 564 months										
r^e_{EW}	15.31	13.49	11.98	11.64	10.83	10.48	10.15	9.06	8.63	7.67	-7.65
	(4.02)	(4.08)	(3.87)	(3.83)	(3.65)	(3.49)	(3.39)	(3.01)	(2.78)	(2.32)	(-4.10)
σ^e_{EW}	26.08	22.69	21.22	20.84	20.36	20.56	20.51	20.66	21.31	22.66	12.79
r^e_{VW}	6.63	7.73	6.86	8.24	7.24	6.36	6.66	4.87	5.68	5.33	-1.29
	(1.85)	(2.40)	(2.44)	(2.94)	(2.67)	(2.47)	(2.65)	(2.01)	(2.29)	(2.08)	(-0.59)
σ^e_{VW}	24.62	22.10	19.26	19.17	18.56	17.65	17.21	16.62	17.01	17.55	15.14
	Expansions, 468 months										
r^e_{EW}	10.21	9.04	7.78	7.73	7.00	6.76	6.60	5.46	5.12	4.75	-5.46
	(2.48)	(2.53)	(2.33)	(2.35)	(2.18)	(2.06)	(2.02)	(1.66)	(1.52)	(1.33)	(-2.72)
σ^e_{EW}	25.67	22.31	20.84	20.57	20.10	20.50	20.38	20.49	21.08	22.27	12.53
r^e_{VW}	3.50	5.07	5.19	5.57	4.95	4.38	4.59	2.84	4.37	4.18	0.67
	(0.90)	(1.47)	(1.71)	(1.83)	(1.68)	(1.53)	(1.69)	(1.07)	(1.65)	(1.53)	(0.27)
σ^e_{VW}	24.33	21.49	18.98	18.98	18.42	17.89	16.96	16.56	16.57	17.10	15.38
	Contractions, 96 months										
r^e_{EW}	40.16	35.20	32.45	30.66	29.48	28.61	27.47	26.61	25.78	21.88	-18.28
	(4.21)	(4.22)	(4.15)	(4.06)	(4.00)	(4.01)	(3.79)	(3.60)	(3.34)	(2.56)	(-3.79)
σ^e_{EW}	26.99	23.57	22.14	21.37	20.85	20.18	20.50	20.88	21.82	24.20	13.65
r^e_{VW}	21.86	20.67	15.00	21.23	18.39	15.98	16.72	14.75	12.08	10.96	-10.89
	(2.41)	(2.37)	(2.07)	(3.04)	(2.74)	(2.78)	(2.60)	(2.50)	(1.80)	(1.58)	(-2.26)
σ^e_{VW}	25.64	24.62	20.54	19.78	18.98	16.24	18.22	16.67	18.98	19.63	13.65

[17] We apply a one-sided HP-filter to the monthly industrial production series (smoothing parameter is 129,600 based on Ravn and Uhlig (2002)). The industrial production for the past month is typically announced 2-3 weeks after the end of the month. We designate recession/expansion years at the end of June of each year based on the industrial production levels in May to make sure that the data is available to investors in real time. NBER business cycle dates are typically announced several months after the start of recessions/expansions, and so are not available to investors in real time. Nevertheless, our recession designations based on industrial production and NBER dates match almost perfectly, so the results are nearly identical if we use NBER dates instead.

Note: r^e_{EW} is equal-weighted monthly excess returns (excess of risk free rate). r^e_{VW} is value-weighted monthly excess returns, annualized, averages are taken over time (%). σ^e_{EW} and σ^e_{VW} are the corresponding standard deviations. Contemporaneous returns are measured in the year of the portfolio formation, from January of year t to December of year t. Future returns are measured in the year following the portfolio formation, from July of year $t+1$ to June of year $t+2$ and annualized (%). $t-$ statistics are in parentheses. Expansion and contraction periods are designated in June of year $t+1$ based on the level of (one sided HP-filtered) industrial production in May of that year. Returns over the expansions and contractions are measured from July of year $t+1$ to June of year $t+2$.

We find that the negative relationship between TFP and expected returns persists both in expansions and in contractions for equal-weighted portfolios. However, there are significant differences in returns over the business cycles. The average level of expected returns is much higher in recessions, approximately 30%, than expansions, 7%. The spread between the returns of high and low TFP portfolios are also much higher during contractions, 18.3%, than expansions, 5.5%, in equal weighted portfolios. For value-weighted portfolios, the spread is 11% and is significant over contractions. During expansions, however, the spread is not significant.

Even though all portfolios have higher expected returns in recessions, the increase in expected returns of low TFP portfolios is roughly twice as much as it is for the high TFP portfolios. We interpret the spread in the average returns across these portfolios, especially in recessions, as the risk premia associated with the higher risk of low productivity firms.

In Table V we investigate whether widely used asset pricing models such as the CAPM and Fama-French 3-factor model capture the variation in excess returns of TFP sorted portfolios. As we demonstrate in Table III, TFP is significantly related to size and B/M at the firm level. Hence, we explore whether the returns of TFP sorted portfolios are systematically related to SMB and HML.[18]

Table V presents the alphas and betas of TFP sorted portfolios for the CAPM and FF models. Betas are estimated by regressing the portfolio excess returns on the factors. Alphas are estimated as intercepts from the regressions of excess portfolio returns. Monthly alphas are annualized by multiplying by 12. The top panel reports the results for the equally-weighted portfolios, and the lower panel reports the value-weighted portfolio results.

[18] MKT is excess market returns; SMB is returns of portfolio that is long in small, short in big firms; HML is returns of portfolio that is long in high B/M, short in low B/M firms. (Fama and French, 1992, 1993, among others)

Table V: Alphas and Betas of Portfolios Sorted on TFP (%, annualized)
Dependent Variable: Excess Returns, July 1964 - June 2011

	Low	2	3	4	5	6	7	8	9	High	High-Low
						Equal-weighted Portfolios					
						CAPM					
alpha	8.86	7.52	6.17	5.79	5.04	4.54	4.11	2.92	2.28	0.88	-7.99
	(3.50)	(3.75)	(3.55)	(3.59)	(3.32)	(3.10)	(3.03)	(2.21)	(1.70)	(0.63)	(-4.27)
MKT	1.24	1.15	1.11	1.12	1.11	1.14	1.16	1.18	1.22	1.30	0.07
	(26.79)	(31.32)	(35.16)	(38.19)	(40.16)	(42.77)	(46.93)	(49.15)	(49.97)	(51.21)	(1.93)
						FF					
alpha	4.37	2.60	1.67	1.25	0.87	0.79	1.13	0.30	0.28	-0.02	-4.39
	(2.62)	(2.09)	(1.61)	(1.42)	(0.99)	(0.89)	(1.34)	(0.35)	(0.32)	(-0.02)	(-2.74)
MKT	1.02	1.02	1.01	1.03	1.03	1.05	1.06	1.08	1.09	1.14	0.12
	(31.33)	(42.25)	(49.95)	(60.46)	(60.20)	(60.49)	(64.34)	(63.86)	(63.34)	(63.30)	(3.79)
HML	0.27	0.46	0.44	0.47	0.43	0.37	0.24	0.19	0.08	-0.12	-0.39
	(5.56)	(12.70)	(14.38)	(18.10)	(16.63)	(13.96)	(9.83)	(7.66)	(2.94)	(-4.48)	(-8.38)
SMB	1.27	1.01	0.88	0.84	0.77	0.73	0.69	0.66	0.67	0.66	-0.61
	(27.89)	(29.98)	(31.19)	(35.16)	(32.16)	(29.99)	(30.15)	(27.78)	(27.77)	(26.40)	(-13.87)
						Value-weighted Portfolios					
						CAPM					
alpha	-0.19	1.20	1.16	2.39	1.65	0.92	1.33	-0.34	0.40	-0.09	0.10
	(-0.10)	(0.83)	(0.93)	(2.21)	(1.48)	(0.99)	(1.52)	(-0.43)	(0.47)	(-0.10)	(0.05)
MKT	1.31	1.25	1.09	1.12	1.07	1.04	1.02	1.00	1.01	1.04	-0.27
	(36.60)	(47.34)	(47.94)	(56.80)	(53.09)	(61.42)	(63.92)	(70.89)	(65.41)	(63.12)	(-6.91)
						FF					
alpha	-1.53	-0.60	-0.57	0.50	-0.13	0.23	0.91	-0.22	1.44	1.82	3.35
	(-0.88)	(-0.43)	(-0.47)	(0.51)	(-0.12)	(0.24)	(1.02)	(-0.28)	(1.77)	(2.25)	(1.87)
MKT	1.17	1.23	1.10	1.12	1.11	1.07	1.04	1.01	1.03	1.02	-0.15
	(34.66)	(45.45)	(46.76)	(57.59)	(54.49)	(59.04)	(60.73)	(66.17)	(64.93)	(64.58)	(-4.30)
HML	-0.03	0.21	0.23	0.24	0.29	0.13	0.09	0.00	-0.11	-0.28	-0.26
	(-0.57)	(5.11)	(6.45)	(8.22)	(9.36)	(4.70)	(3.35)	(0.02)	(-4.81)	(-11.97)	(-4.85)
SMB	0.63	0.28	0.20	0.24	0.08	0.00	-0.02	-0.05	-0.17	-0.15	-0.78
	(13.46)	(7.30)	(6.13)	(8.87)	(2.78)	(-0.19)	(-0.94)	(-2.32)	(-7.84)	(-6.73)	(-16.03)

Note: The table presents the regressions of equal-weighted and value-weighted excess portfolio returns on various factor returns. MKT, SMB, and HML factors are taken from Ken French's website. The portfolios are sorted on TFP. Alphas are annualized (%). Returns are measured from July 1964 to June 2011. $t-$statistics are calculated by dividing the slope coefficient by its time series standard error and presented in parentheses.

We find that low TFP portfolios load heavily on SMB, whereas the loadings of the high TFP portfolios are low, even negative in some cases. The loadings on HML are non-monotonic, but higher TFP portfolios have lower loadings than lower TFP portfolios. The equal-weighted portfolios have a non-monotonic loading on MKT, whereas the value-weighted low TFP portfolios have significantly higher loading on MKT compared to the high TFP portfolios. Neither the CAPM, nor the FF 3-factor model completely explain the return spread in the equally weighted portfolios: High-Low TFP portfolio has a CAPM alpha around –8%, and FF alpha of -4.5%, which are both statistically significant. The spreads in alphas of value-weighted portfolios are positive but not statistically significant.[19]

Our takeaway from these results is not necessarily that TFP is a separate risk factor that is not captured by these factors but rather that TFP is systematically related to SMB, and to some extent to HML, which we further investigate empirically and through a model economy.

1.4.2 Fama-MacBeth Regressions

In Section 1.4.1, we examine the characteristics and realized average returns of portfolios sorted on TFP. In this section, we run Fama-MacBeth cross-sectional regressions (Fama and MacBeth, 1973) of monthly stock returns on lagged firm level TFP as well as other control variables. The estimates of the slope coefficients in Fama-MacBeth regressions allow us to determine the magnitude of the effect of the firm characteristics on excess stock returns.

Table VI reports the time series averages of cross-sectional regression slope coefficients and their time-series t-statistic (computed as in Newey-West with 4 lags) obtained from the Fama-MacBeth regressions. In all specifications, the dependent variable is the excess monthly stock returns, annualized to make the magnitudes comparable to the results in Table IV. The first specification in Table VI shows the relationship between the level of TFP and future excess returns. The cross sectional regression, where log TFP is the only explanatory variable, produces a negative and statistically significant average slope. The magnitude of the effect is significant as well. The -5.66 average regression coefficient in this setting translates into approximately 7.5% higher expected returns for the firms in the lowest TFP decile compared to a firm in the highest TFP decile.

[19] In an untabulated, extended specification, we include two other commonly used factors, Carhart's momentum factor (MOM) and Pastor and Stambaugh's liquidity factor (LIQ). Pastor and Stambaugh (2003) document a strong relationship between size and liquidity, so higher returns of low TFP firms (which are typically smaller) could potentially be explained by their exposure to the liquidity factor. We find that MOM and LIQ factors are not systematically related to the returns of the TFP sorted portfolios, and the results are very similar to that of the FF model.

Table VI: Fama-MacBeth Regressions with Other Predictors (%, annualized)
July 1964 - June 2011

	Int	TFP	BM	SIZE	I/K	HR	INV	AG	ROE	ROA	GPR	NS	RER	LEV	RD/PPE	AGE
I	10.6	-5.7														
	(3.0)	(-3.6)														
II	11.8	-1.0	4.5													
	(3.4)	(-0.8)	(5.0)													
III	7.0	-1.8		-1.9												
	(2.3)	(-1.6)		(-4.1)												
IV	11.8	-4.4			-8.0											
	(3.6)	(-2.6)			(-3.3)											
V	11.0	-4.9				-3.7										
	(3.2)	(-3.0)				(-4.2)										
VI	10.9	-5.2					-2.0									
	(3.2)	(-3.1)					(-3.5)									
VII	11.6	-3.9						-5.5								
	(3.4)	(-2.4)						(-4.9)								
VIII	10.8	-4.3							-4.9							
	(3.0)	(-3.3)							(-2.1)							
IX	11.8	-2.7								-22.2						
	(3.0)	(-2.1)								(-1.9)						
X	5.9	-6.2									5.9					
	(3.4)	(-4.0)									(3.4)					
XI	10.5	-5.6										-1.2				
	(3.1)	(-3.6)										(-1.5)				
XII	11.9	-6.0											1.0			
	(3.2)	(-3.2)											(0.5)			
XIII	8.6	-4.2												7.5		
	(2.7)	(-3.1)												(2.3)		
XIV	10.3	-6.0													6.9	
	(3.1)	(-3.8)													(1.5)	
XV	12.1	-5.8														-0.1
	(2.8)	(-3.7)														(-1.4)

Note: The table presents the average slopes and their $t-$statistics from monthly cross-section regressions to predict excess stock returns. Results are reported for the whole sample period of July 1964 to June 2011. The $t-$statistics for the average regression slopes use the time-series standard deviations of the monthly slopes (computed as in Newey-West with 4 lags) and presented in parentheses. Excess returns are predicted for July of year $t+1$ to June of $t+2$. Average TFP each year is normalized to be 1. SIZE is the log market capitalization of firms in June of year $t+1$. Average size each year is normalized to 1. B/M is the ratio of book equity for the last fiscal year-end in year t divided by market equity in December of year t. I/K is the fixed investment to capital ratio, AG is the change in the natural log of assets, HR is the change in the natural log of number of employees, and INV is the change in the natural log of total inventories, all measured from year $t-1$ to year t. NS is the change in the natural log of the split-adjusted shares outstanding from the fiscal year-end in $t-1$ to t. LEV is the ratio of long-term debt holdings in year t to the firm's total assets calculated as the sum of their long-term debt and the market value of their equity in December of year t. RER is the ratio of buildings+capital leases to PPE in year t , adjusted for industries. ROE is the net income in year t divided by book equity for year t. ROA is the net income in year t divided by total assets for year t. GPR is the gross profits in year t divided by book assets for year t. RD/PPE is the ratio of research and development expenses to gross PPE in year t. AGE is computed in year t as the number of years since the firm first shows up in Compustat. Excess returns are annualized (%).

In specifications II-XV, we examine the marginal predictive power of TFP after controlling for several firm level characteristics that are known to predict stock returns and/or found to systematically vary with TFP in Table III.

Specification II considers firm's B/M, III considers size, IV considers investment to capital ratio, V considers hiring rate, VI considers inventory growth, and VII considers asset growth. Specifications VIII through X consider different measures of profitability: ROE in VIII, ROA in IX, and gross profitability in specification X. We study net share issues in XI, real estate ratio in XII, leverage in XIII, R&D/PPE in XIV, and firm age in XV. The definitions of these variables are available in Appendix, Section 4.2. We observe that the cross sectional regressions in specifications I to XV produce negative and statistically significant average slopes for TFP except for the specifications in II and III where including book to market (in II) and size (in III) erodes the significance of TFP.

The results of these predictability regressions should be interpreted with caution, however. In production based asset pricing models, shocks to firm level productivity create ex-post differences across firms in terms of the level of TFP, which affect the riskiness of a stock and its future expected returns. The same shock is the fundamental variable that drives the differences in size, B/M, and many other firm characteristics across firms. To the extent that such models correctly describe firm behavior, characteristics such as size, B/M, and investment/capital ratio should not have additional explanatory power vis-a-vis productivity but should be determined by it. Furthermore, even if firm productivity is the fundamental variable that drives firm behavior, it is arguably measured with more noise than characteristics such as size or B/M, which are observable rather than estimated. As a result, in return predictability regressions which include both firm productivity and other firm characteristics as forecasting variables, the better measured variables are expected to drive out productivity. Our overall goal is to shed some light on why firm characteristics can rationally predict returns and not necessarily to come up with a variable that outperforms other firm characteristics in return predictability regressions.

TFP, Expected Profitability, and Expected Investment The results of the Fama-MacBeth regressions, similar to our earlier results based on portfolio sorts, demonstrate that high TFP firms have lower future returns. In section 1.3 we also document that TFP is related to many other variables, in particular, several variables related to firm investments (such as I/K, asset growth, inventory investment) and profits (ROE, ROA, gross profitability), that are known to predict returns. Specifically, we find that TFP is positively related to both firm investments and profitability. Thus our results are somewhat surprising, given that the literature has documented negative relationship between investments and future returns, but generally positive relationship between profitability and future returns.

Fama and French (2006) uses the basic valuation equation to demonstrate the basic relationship between the market/book value of the equity, expected future profitability, expected future investments, and expected returns:[20]

$$\frac{M_t}{B_t} = \sum_{\tau=1}^{\infty} \frac{E\left(\frac{Y_{t+\tau}}{B_t} - \frac{\Delta B_{t+\tau}}{B_t}\right)}{(1+r)^{\tau}} \tag{3}$$

where M_t and B_t are the market and book value of equity, respectively, and Y_t is the equity earnings per share. The predictions that come out of the valuation equation about the link between expected returns, r, expected investments, $E_t(\frac{\Delta B_{t+\tau}}{B_t})$, and expected profitability, $E_t\left(\frac{Y_{t+\tau}}{B_t}\right)$ are: (1) Controlling for the M/B, and expected profitability, firms with high expected investment (and asset growth) have low expected stock returns. (2) Controlling for the M/B, and expected asset growth, firms with high expected profitability have high expected stock returns. Therefore, expected profitability and expected investment are related to expected returns with the opposite sign after controlling for the remaining two variables. Proxying for expected future profitability and investments with current profitability and investments, Fama and French (2006) have confirmed these relationships in the data.[21]

Table VII investigates how TFP is related to future profitability and asset growth in equation 3.[22] We find that TFP strongly predicts both future profitability and future asset growth for several years. The slope coefficient estimates imply that profitability and asset growth of firms

[20] Recently, Hou, Xue and Zhang (2012) derive similar relationships using a simple two-period investment based asset pricing model.

[21] Novy-Marx (Forthcoming) and Hou, Xue and Zhang (2012) report similar findings.

[22] We scale firm profits by total assets, rather than book equity, to be able to compare the slope coefficients from two regressions.

in the highest TFP decile will be roughly 15 and 20 percentage points higher, respectively, than the firms in the lowest TFP decile in the next year. Slightly higher slope coefficient estimates for the asset growth, at least in the first year, imply that TFP produces wider spreads in future asset growth than in future profitability.[23] Furthermore, TFP is strongly positively related to M/B (Table III), which, according to equation 3, implies low expected returns. Thus our results appear consistent with basic valuation theory and the recent empirical work in this area.

Table VII: Forecasting Profitability and Asset Growth with TFP

	TFP slope coefficients predicting:	
j (years)	$\frac{\text{Profit}_{t+j}}{\text{Asset}_t}$	$\frac{\text{Asset}_{t+j}-\text{Asset}_{t+j-1}}{\text{Asset}_t}$
1	0.107	0.154
	(20.2)	(11.3)
2	0.098	0.111
	(14.4)	(5.26)
3	0.096	0.087
	(9.42)	(3.43)

Note: The table presents the average slopes and their $t-$statistics from annual cross-section regressions to predict future profitability ($\frac{\text{Profit}_{t+j}}{\text{Asset}_t}$) and asset growth ($\frac{\text{Asset}_{t+j}-\text{Asset}_{t+j-1}}{\text{Asset}_t}$). Results are reported for the whole sample period of 1963 to 2009. The $t-$statistics for the average regression slopes use the time-series standard deviations of the annual slopes (computed as in Newey-West with 4 lags) and presented in parentheses. The independent variable is TFP. Profit is net income, as defined in Fama and French (2008). Asset is total assets.

1.4.3 Ex-Ante Discount Rates of TFP-Sorted Portfolios

In both the portfolio approach and the Fama-MacBeth regressions reported in the previous section, we proxy expected returns with ex-post realized returns. A common concern about approximating expected returns with realized returns is that the realized returns are very volatile and can be a bad proxy for expected returns, especially with relatively short time series data.

[23] Fama and French (2006) also document that profitability and asset growth are related. They find that, in univariate regressions, current profitability predicts future asset growth and current asset growth predicts future profitability with positive sign. In untabulated results, we find that TFP predicts both variables better than they predict each other. Also, sorting firms on TFP produces wider spreads in asset growth compared to the spreads generated by sorting on profitability; and also wider spreads in profitability, compared to the spreads generated by sorting firms on asset growth. Hence, TFP is more related to asset growth and profitability, than they are to each other.

To address this concern, we use an ex-ante measure of the discount rates, the implied cost of capital, and examine its cross-sectional relationship with TFP.

The implied cost of capital (ICC) of a given firm is the internal rate of return that equates the firm's stock price to the present value of expected future cash flows (earnings forecasts). Most ICC estimates in the literature, such as Gebhardt, Lee, and Swaminathan (GLS, 2001), rely on analyst forecasts of future earnings. However, analyst forecasts are not available in the first few years of our sample period. Furthermore, analysts tend to follow larger and more visible stocks, so earnings forecasts of many firms in our sample are not available. As an alternative, Hou, van Dijk, and Zhang (HVDZ, 2012) and Wu and Zhang (WZ, 2012) use statistical models to forecast earnings and ROE, respectively. We use ICC measures calculated as described in GLS, HVDZ, and WZ.

ICC for each firm is estimated at the end of June of each calendar year t using the end-of-June firm market value and the earnings and ROE forecasts made at the previous fiscal year end. We match the ICC estimates of individual firms with these firms' most recent TFP estimates. The summary statistics for ICCs of the matched sample are reported in Table VIII. HVDZ and WZ based ICCs span July 1970 - June 2011 period, with roughly 6,800+ unique firms and 63,000 firm $\times$ year observations. GLS based ICCs start in July 1976, and have roughly 6,250 unique firms and 47,000 observations.

Table VIII: Implied Cost of Capital Summary Statistics

	Sample Period	Number of Firms	Number of Observations	Mean (%, annualized)	Median (%, annualized)	Std Dev (%, annualized)
GLS	July 1976-June 2011	6249	46,873	9.42	9.02	5.54
HVDZ	July 1970-June 2011	6886	63,030	10.82	9.30	8.88
WZ	July 1970-June 2011	6826	62,556	8.67	8.19	4.23

Note: The table presents the summary statistics for various measures of implied cost of capital (ICC) after they are matched to our TFP data. ICC data based on the Gebhardt, Lee, Swaminathan (2001) method (GLS) is matched to the TFP data from 1975 to 2009. ICC data based on the Hou, van Dijk and Zhang (2012) method (HVDZ) is matched to the TFP data from 1969 to 2009. ICC data based on the Wu and Zhang (2012) method (WZ) is matched to the TFP data from 1969 to 2009. The number of firms is the number of distinct firms, and the number of observations is the number of firm $\times$ year observations. The means, medians, and standard deviations are calculated by pooling all observations based on each method.

Table IX: Implied Cost of Capital for TFP Sorted Portfolios (%, annualized)

	Low	2	3	4	5	6	7	8	9	High	High-Low
					All states						
GLS_{EW}	10.43	10.98	10.84	10.59	10.40	10.03	9.68	9.41	9.08	8.96	-1.47
											(-6.89)
GLS_{VW}	9.15	9.66	9.67	9.48	9.27	8.98	8.79	8.43	8.04	8.33	-0.82
											(-3.57)
$HVDZ_{EW}$	13.81	14.44	13.38	12.36	11.47	10.66	10.10	9.49	8.82	8.37	-5.45
											(-18.46)
$HVDZ_{VW}$	8.33	9.23	9.10	8.93	8.52	8.26	7.98	7.53	7.04	7.04	-1.29
											(-4.30)
WZ_{EW}	9.06	10.28	10.18	9.95	9.61	9.25	8.78	8.37	7.80	7.25	-1.81
											(-7.13)
WZ_{VW}	8.07	8.95	9.03	8.87	8.55	8.38	8.03	7.62	7.16	6.86	-1.22
											(-4.12)
					Expansions						
GLS_{EW}	10.24	10.72	10.63	10.19	10.27	9.85	9.52	9.31	8.98	8.85	-1.39
											(-5.86)
GLS_{VW}	8.93	9.43	9.51	9.21	9.16	8.83	8.58	8.30	7.93	8.20	-0.72
											(-2.71)
$HVDZ_{EW}$	13.12	13.65	12.81	11.86	11.07	10.32	9.74	9.17	8.55	8.10	-5.02
											(-15.63)
$HVDZ_{VW}$	7.89	8.90	8.68	8.59	8.21	8.04	7.71	7.29	6.87	6.79	-1.10
											(-3.43)
WZ_{EW}	8.71	9.89	9.85	9.63	9.33	9.01	8.53	8.11	7.59	7.07	-1.64
											(-5.91)
WZ_{VW}	7.78	8.65	8.72	8.59	8.32	8.16	7.84	7.41	7.08	6.73	-1.05
											(-3.10)
					Contractions						
GLS_{EW}	11.32	12.22	11.84	12.53	11.00	10.91	10.48	9.90	9.58	9.47	-1.85
											(-3.76)
GLS_{VW}	10.24	10.78	10.44	10.78	9.78	9.74	9.80	9.04	8.58	8.95	-1.30
											(-4.26)
$HVDZ_{EW}$	16.64	17.67	15.74	14.41	13.12	12.04	11.60	10.79	9.95	9.45	-7.20
											(-30.39)
$HVDZ_{VW}$	10.14	10.62	10.84	10.35	9.80	9.15	9.07	8.51	7.73	8.06	-2.08
											(-2.87)
WZ_{EW}	10.50	11.93	11.53	11.26	10.75	10.24	9.80	9.43	8.67	8.00	-2.50
											(-4.26)
WZ_{VW}	9.28	10.17	10.27	10.03	9.47	9.26	8.85	8.49	7.53	7.39	-1.89
											(-3.69)

Note: GLS_{EW}, $HVDZ_{EW}$, and WZ_{EW} are equal-weighted implied cost of capital; GLS_{VW}, $HVDZ_{VW}$, and WZ_{VW} are value-weighted implied cost of capital, annual, averages are taken over time (%). ICC data based on the Gebhardt, Lee, Swaminathan (2001) method (GLS) is matched to the TFP data from 1975 to 2009. ICC data based on Hou, van Dijk, and Zhang (2012) method (HVDZ) is matched to the TFP data from 1969 to 2009. ICC data based on Wu and Zhang (2012) method (WZ) is matched to the TFP data from 1969 to 2009. HVDZ and WZ expected returns are estimated for the July 1970-June 2011 period, GLS expected returns start in July 1976. $t-$ statistics are in parentheses. Expansion and contraction periods are designated in June of each year (when ICCs are computed) based on the level of (one sided HP-filtered) industrial production in May of that year. Returns over the expansions and contractions are measured from July of that year to June of the following year.

Table IX presents the average implied cost of capital estimates for portfolios sorted on productivity. The relationship between TFP and the cost of capital measured from all three ICC measures is negative and quite monotonic. The firms with low productivity have higher discount rates (ICC) than firms with high productivity using all ICC measures and both for equal and value weighted portfolios. The spreads between the high and low TFP firms' expected returns are all negative and highly statistically significant, implying that firms with low productivity have higher ex-ante discount rates, and so are riskier than high productivity firms.

In addition to the unconditional discount rates, Table IX also presents the ICCs conditional on whether the economy is expanding or contracting around the portfolio formation time. Similar to the results based on average realized returns in Table IV, both the levels of implied cost of capital and the spread between the low and high TFP portfolios are countercyclical. The spreads between the discount rates of low and high TFP portfolios increase roughly 50-100% for all ICC measures and for both equal and value-weighted portfolios as the economy moves from expansions to contractions. However, most of the increases in expected returns happen in the low TFP portfolio returns. Even though all firms are riskier in recessions, firms with low productivity are hit particularly hard and thus bear more risk than firms with high TFPs.

1.4.4 Inspecting the Mechanism with Operating Performance

We have shown that firm level TFPs are systematically related to many firm characteristics and firms with low TFP have higher future realized returns and implied cost of capital. Furthermore, we document that the return spreads between the low and high TFP firms increase during recessions, and this increase is mostly due to a surge in expected returns of low TFP firms. Our interpretation of this return evidence is that low TFP firms are riskier than high TFP firms, and they are particularly risky in recessions. In this section, we provide additional evidence on the higher risk of low TFP firms that is not based on realized or ex-ante returns. We investigate whether there are systematic differences in the sensitivity of low and high TFP firms' operating performance to aggregate shocks in the economy. The existence of such differences would shed some light on the mechanism behind the risk and return differentials between low and high TFP firms.

We conjecture that operating performance of low TFP firms would be more sensitive to aggregate shocks than the high TFP firms, especially during bad economic times. We measure the operating performance of firms with their profitability. In order to test this hypothesis, we examine pooled time series/cross sectional regressions of the form:

$$\Delta\text{Profit}_{it} = b_0 + b_1\Delta\text{Profit}_{agg,t} + b_2\Delta\text{Profit}_{agg,t} \times IP_{t-1} + \varepsilon_{it} \tag{4}$$

where $\Delta\text{Profit}_{i,t}$ is the change in the profits (net income) of firm i between year $t-1$ and t, scaled by firm's assets in year $t-1$.[24] We proxy aggregate shocks with the cross sectional average of $\Delta\text{Profit}_{i,t}$ over all firms in our sample.[25] Our business cycle variable is the one sided HP-filtered industrial production, defined earlier in Section 1.4.1, standardized to have a standard deviation of 1. Industrial production is measured in May of year $t-1$ (released in June). We separate firms into 10 TFP groups based on their TFPs in year $t-1$, similar to the decile portfolios used earlier. We run panel regressions for in each TFP group.

Table X reports the results of the profitability regressions. Examination of Panel A reveals that, unconditionally, low TFP firms have higher sensitivity to aggregate shocks in the economy. The regression coefficient is 1.5 for the firms in the lowest TFP group, goes down to around 0.8 for higher TFP groups, and increases to 1.1 for the firms in the highest TFP group. More importantly, Panel B shows that low TFP firms' sensitivity to aggregate shocks increases in bad economic times (when industrial production is low). One standard deviation decrease in industrial production increases the sensitivity of lowest TFP firms by approximately 0.5. The pattern reverses for high TFP firms: Their profits' sensitivity to aggregate shocks increases in good times by approximately 0.3. These results are consistent with our earlier findings based on returns/discount rates and imply a higher risk for low TFP firms, especially when the economy is doing poorly.

[24] This regression is obtained from writing our empirical model in levels with year and firm fixed effects and then taking the first differences, which leads to the elimination of year and firm fixed effects.

[25] Replacing $\Delta\text{Profit}_{agg,t}$ with change in aggregate labor productivity reported by the BLS yields similar results. Note that we cannot calculate aggregate productivity by averaging our firm level TFPs as our TFP estimates are free of industry and year effects. Using ΔProfit on each side of the regression, at the firm level on the LHS and aggregate on the RHS, brings a beta-like interpretation to the regression coefficients.

Table X: Profitability Regressions for TFP Sorted Panels
Dependent Variable: $\Delta\text{Profit}_{i,t}$

	Low TFP	2	3	4	5	6	7	8	9	High TFP
Panel A										
intercept	0.04	0.00	0.00	-0.01	-0.01	-0.01	0.00	-0.01	-0.01	-0.01
	(7.47)	(0.44)	(0.32)	(-6.41)	(-6.01)	(-3.97)	(-2.60)	(-4.27)	(-2.17)	(-1.92)
$\Delta\text{Profit}_{agg,t}$	1.53	1.34	0.81	1.01	1.01	0.93	0.66	0.82	0.86	1.09
	(3.52)	(8.04)	(7.15)	(10.89)	(9.75)	(6.02)	(8.26)	(7.35)	(5.00)	(4.00)
Panel B										
intercept	0.05	0.00	0.00	-0.01	-0.01	-0.01	0.00	-0.01	-0.01	-0.01
	(7.38)	(1.15)	(0.38)	(-6.81)	(-5.67)	(-5.17)	(-2.96)	(-3.83)	(-2.13)	(-2.03)
$\Delta\text{Profit}_{agg,t}$	1.11	1.05	0.77	0.95	0.95	1.06	0.74	0.99	0.99	1.35
	(2.70)	(9.27)	(6.10)	(12.58)	(10.13)	(9.98)	(12.14)	(8.66)	(4.96)	(4.98)
$\Delta\text{Profit}_{agg,t}$	-0.56	-0.38	-0.04	-0.08	-0.07	0.17	0.10	0.23	0.17	0.33
$\times IP_{t-1}$	(-2.95)	(-4.62)	(-0.88)	(-1.26)	(-1.05)	(1.83)	(2.09)	(3.78)	(2.82)	(2.93)

Note: The table presents the results of panel regressions of change in firm level profitability on change in aggregate profitability and change in aggregate profitability conditional on past industrial production. Change in profitability is measured as the level difference between current and past profits, scaled by lagged assets. Change in aggregate profitability is measured as the cross sectional average of firm level changes. Industrial production is the level of (one sided HP-filtered) industrial production in May of the past year, standardized to have a standard deviation of 1. Firms are sorted into 10 equal-sized TFP groups based on past year's TFPs. The sample period is 1964-2009, total number of observations is 78,382. Standard errors are clustered by date and $t-$ statistics are in parentheses.

2 Model

In this section we investigate whether a model where firms are subject to both aggregate and idiosyncratic productivity shocks is capable of accounting for the cross sectional relationship between TFP, firm level characteristics, and stock returns documented in Tables III, IV, and V. We calibrate the model using the firm level TFP estimates summarized in Section 1.2 and examine the resulting firm level characteristics and firm returns generated by the model economy.

2.1 Firms

There are many firms that produce a homogeneous good using capital and labor. These firms are subject to different productivity shocks.

The production function for firm i is given by:

$$\begin{aligned} Y_{it} &= F(A_t, Z_{it}, K_{it}, L_{it}) \\ &= A_t Z_{it} K_{it}^{\alpha_k} L_{it}^{\alpha_l}. \end{aligned}$$

K_{it} denotes the beginning of period t capital stock of firm i. L_{it} denotes the labor used in production by firm i during period t. Labor and capital shares are given by α_l and α_k where $\alpha_l + \alpha_k \in (0, 1)$. Aggregate productivity is denoted by $a_t = \log(A_t)$. a_t has a stationary and monotone Markov transition function, given by $p_a(a_{t+1}|a_t)$, as follows:

$$a_{t+1} = \rho_a a_t + \varepsilon_{t+1}^a \tag{5}$$

where $\varepsilon_{t+1}^a \sim$ i.i.d. $\mathrm{N}(0, \sigma_a^2)$. The firm productivity, $z_{it} = \log(Z_{it})$, has a stationary and monotone Markov transition function, denoted by $p_{z_i}(z_{i,t+1}|z_{it})$, as follows:

$$z_{i,t+1} = \rho_z z_{it} + \varepsilon_{i,t+1}^z \tag{6}$$

where $\varepsilon_{i,t+1}^z \sim$ i.i.d. $\mathrm{N}(0, \sigma_z^2)$. $\varepsilon_{i,t+1}^z$ and $\varepsilon_{j,t+1}^z$ are uncorrelated for any pair of firms (i, j) with $i \neq j$.

The capital accumulation rule is:

$$K_{i,t+1} = (1 - \delta)K_{it} + I_{it}$$

where I_{it} denotes investment and δ denotes the depreciation rate of installed capital.

Investment is subject to quadratic adjustment costs given by g_{it}:

$$g(I_{it}, K_{it}) = \frac{1}{2}\eta \left(\frac{I_{it}}{K_{it}} - \delta\right)^2 K_{it} \tag{7}$$

with $\eta > 0$. In this specification, investors incur no adjustment cost when net investment is zero, i.e., when the firm replaces its depleted capital stock and maintains its capital level.

Firms are equity financed and face a perfectly elastic supply of labor at a given stochastic equilibrium real wage rate W_t as in Bazdresch, Belo, and Lin (2010) and Jones and Tüzel (Forthcoming). The equilibrium wage rate, given by:

$$W_t = exp(\omega a_t) \tag{8}$$

is assumed to be increasing with aggregate productivity, with $0 < \omega < 1$ determining the sensitivity of the relation.[26] Hiring decisions are made after firms observe the productivity shocks and labor is adjusted freely; hence, for each firm, marginal product of labor equals the wage rate:

$$\begin{aligned} F_{L_{it}} &= F_L(A_t, Z_{it}, K_{it}, L_{it}) \\ &= W_t. \end{aligned}$$

Dividends to shareholders are equal to:

$$D_{it} = Y_{it} - [I_{it} + g_{it}] - W_t L_{it}. \tag{9}$$

At each date t, firms choose $\{I_{i,t}, L_{i,t}\}$ to maximize the net present value of their expected dividend stream, which is the firm value:

$$V_{it} = \max_{\{I_{i,t+k}, L_{i,t+k}\}} E_t \left[\sum_{k=0}^{\infty} M_{t,t+k} D_{i,t+k} \right], \tag{10}$$

subject to (Eq.5-8), where $M_{t,t+k}$ is the stochastic discount factor between time t and $t+k$. V_{it} is the cum-dividend value of the firm.

The first order conditions for the firm's optimization problem leads to the pricing equation:

$$1 = \int \int M_{t,t+1} R^I_{i,t+1} p_{z_i}(z_{i,t+1}|z_{it}) p_a(a_{t+1}|a_t) d_{z_i} d_a \tag{11}$$

[26] With competitive labor markets, wage equals the value of the marginal product of labor. In general equilibrium with a representative firm, the marginal product of labor is $mpl = exp(at)\alpha_l L^{\alpha_l - 1} K^{\alpha_k}$. Hence, wages are a function of aggregate productivity, total labor force, and the capital stock used in production. Here we abstract from aggregate labor and capital and model wages as a function of aggregate productivity.

where the returns to investment are given by:

$$R^I_{i,t+1} = \frac{F_{K_{i,t+1}} + (1-\delta)q_{i,t+1} + \frac{1}{2}\eta\left(\left(\frac{I_{i,t+1}}{K_{i,t+1}}\right)^2 - \delta^2\right)}{q_{it}} \tag{12}$$

and where

$$F_{K_{it}} = F_K(A_t, Z_{it}, K_{it}, L_{it}).$$

Tobin's q, the consumption good value of a newly installed unit of capital, is:

$$q_{it} = 1 + \eta\left(\frac{I_{it}}{K_{it}} - \delta\right). \tag{13}$$

The pricing equation (Eq.11) establishes a link between the marginal cost and benefit of investing. The term in the denominator of the right hand side of the equation, q_{it}, measures the marginal cost of investing. The terms in the numerator represent the discounted marginal benefit of investing. The firm optimally chooses I_{it} such that the marginal cost of investing equals the discounted marginal benefit.

The returns to the firm are defined as:[27]

$$R^S_{i,t+1} = \frac{V_{i,t+1}}{V_{it} - D_{it}}. \tag{14}$$

2.2 The Stochastic Discount Factor

Since the purpose of our model is to examine the cross sectional variation across firms, we use a framework where time series properties of returns are matched by using an exogenous pricing kernel. Following Berk, Green, and Naik (1999), Zhang (2005), and Gomes and Schmid (2010), we directly parameterize the pricing kernel without explicitly modeling the consumer's problem. As in Jones and Tüzel (Forthcoming), the pricing kernel is given by:

$$\begin{aligned} \log M_{t+1} &= \log\beta - \gamma_t \epsilon^a_{t+1} - \frac{1}{2}\gamma_t^2\sigma_a^2 \\ \log\gamma_t &= \gamma_0 + \gamma_1 a_t \end{aligned} \tag{15}$$

[27] We do not assume constant returns to scale in the production function; i.e.,$\alpha_l + \alpha_k \in (0,1)$. In the presence of constant returns to scale, firm return would be equivalent to the returns to fixed investment, R^I_{t+1}. With slightly decreasing returns to scale, firm returns slightly diverge from the investment returns.

where β, $\gamma_0 > 0$, and $\gamma_1 < 0$ are constant parameters.

This pricing kernel shares a number of similarities with Zhang (2005). M_{t+1}, the stochastic discount factor from time t to $t+1$, is driven by ϵ^a_{t+1}, the shock to the aggregate productivity process in period $t+1$. The volatility of M_{t+1} is time-varying, driven by the γ_t process. This volatility takes higher values following business cycle contractions and lower values following expansions, implying a countercyclical price of risk as the result.[28] In the absence of countercyclical price of risk, the risk premia generated in the economy does not change with economic conditions. Empirically, existence of time varying risk premia is well documented (Fama and Schwert (1977); Fama and Bliss (1987); Fama and French (1989); Campbell and Shiller (1991); Cochrane and Piazzesi (2005); Jones and Tüzel (2013); among many others). Our empirical results in Table IV and Table IX, that average (future) realized returns and implied cost of capital are much higher in contractions, compared to expansions, provide additional motivation for modeling countercyclical price of risk.

2.3 Calibration

Solving our model generates firms' investment and hiring decisions as functions of the state variables, which are the aggregate and firm level productivity and the capital of the firm. Since the stochastic discount factor and the wages are specified exogenously, the solution does not require aggregation. Hence, the distribution of the capital stock, a high dimensional object is not a part of the state space. This feature of the model simplifies the solution of the model significantly.

To be consistent with our annual empirical results, we calibrate the model at annual frequency. Table XI presents the parameters used in the calibration.[29] We derive the parameters of the firm level productivity process from the production function estimations in Section 1.2. The persistence of the firm productivity process, ρ_z, is 0.7. The conditional volatility of firm productivity, σ_z, is computed from ρ_z and the cross sectional standard deviation of firm pro-

[28] A countercyclical price of risk is endogenously derived in Campbell and Cochrane (1999) from time varying risk aversion; in Barberis, Huang, and Santos (2001) from loss aversion; in Constantinides and Duffie (1996) from time varying cross sectional distribution of labor income; in Guvenen (2009) from limited participation; in Bansal and Yaron (2004) from time varying economic uncertainty; and in Piazzesi, Schneider, and Tüzel (2007) from time varying consumption composition risk.

[29] Calibration of the model economy based on alternative TFP estimations (summarized in the Appendix) yields very similar asset pricing results.

ductivity as 0.268 $\left(= 0.38 \times \sqrt{1 - 0.7^2}\right)$.[30] The parameters of the production function, α_k and α_l, are roughly equal to our estimates presented in Table I and Figure 1. Even though our production function estimates imply almost constant returns to scale production technology, we model technology as slightly decreasing returns to scale, with $\alpha_k + \alpha_l = 0.95$. It is well known that firm size is indeterminate with constant returns to scale technology. Decreasing returns to scale technology makes studying the relationship between firm size and productivity possible.

Table XI: Model Parameter Values

Parameter	Description	Value
α_k	Capital share	0.22
α_l	Labor share	0.73
ω	Wage sensitivity to aggregate productivity	0.20
β	Discount factor	0.988
γ_0	Constant price of risk parameter	3.27
γ_1	Time varying price of risk parameter	-13.32
δ	Capital depreciation rate	0.08
η	Adjustment cost parameter	4.45
ρ_a	Persistence of aggregate productivity	0.922
σ_a	Conditional volatility of aggregate productivity	0.014
ρ_z	Persistence of firm productivity	0.7
σ_z	Conditional volatility of firm productivity	0.268

We take the parameters of the aggregate productivity from King and Rebelo (1999) and annualize them. Their point estimates for ρ_a and σ_a are 0.979 and 0.0072, respectively, using quarterly data, implying annual parameters of 0.922 and 0.014. The depreciation rate for fixed capital is set to 8% annually, which is roughly the midpoint of values used in other studies. Cooley and Prescott (1995) use 1.6%; Boldrin, Christiano, and Fisher (2001) use 2.1%; and Kydland and Prescott (1982) use a 2.5% quarterly depreciation rate. We follow Bazdresch, Belo, and Lin (2010) in choosing the wage function parameter ω to equal 0.2, which matches the aggregate wage cyclicality estimated by Merz and Yashiv (2007).

We choose the pricing kernel parameters β, γ_0, and γ_1 to match the average riskless rate and the first two moments of aggregate value-weighted excess stock returns measured from the data used in our empirical exercise. The discount factor β is 0.988, which implies an annual risk free rate of 1.2%. γ_0 and γ_1 are 3.27 and -13.32, respectively, and generate annual excess mean returns and standard deviation of 6.16% and 17%, respectively. Finally, the adjustment

[30] The dispersion of firm productivity rises over time, from 0.23 in 1963 to 0.53 in 2009. In this calibration, we take the cross sectional standard deviation to be 0.375, which is the average dispersion over this time period.

cost parameter, η, is set to 4.45 to replicate the value-weighted average (annual) volatility of investment to capital ratio of 15.7% in our data.

In order to compute the model statistics, we perform 500 simulations of the model economy with 1,000 firms over 50 periods (years), which is roughly comparable to the length and average breadth of our empirical sample.

2.4 Model Results

The key mechanism relating firm level productivity to expected returns involves the interaction of convex adjustment costs and the countercyclical Sharpe ratios assumed in our pricing kernel. Firm risk derives from its inability to freely adjust its capital following shocks to aggregate and firm level productivity. In this economy, aggregate productivity shocks drive the business cycles. In bad times (low aggregate productivity level), net present value of investments go down due to lower expected cash flows and higher discount rates. Hence, all firms would like to invest less and hire less. Even though firms can freely adjust their labor, they incur adjustment costs when they change their capital stock.[31] In states of low aggregate productivity, a bad aggregate shock has a larger negative effect on the low TFP firms. These firms would have a lower investment rate (i.e., reduce their capital stock relatively more) than the high TFP firms. Due to the convexity in adjustment costs, low TFP firms, who are at a steeper part of the adjustment costs, sustain a higher cost while reducing their capital stock in a recession than high TFP firms. A positive aggregate shock, on the other hand, results in a bigger decrease in adjustment costs for these firms compared to more productive firms, hence benefiting the low TFP firms more than the others. Therefore, the returns of the low TFP firms covary more with changes in the economic conditions during recessions. The opposite happens in expansions.[32]

The countercyclical Sharpe ratio breaks the symmetry between recessions and expansions.[33]

[31] Note that adjustment costs are defined over net investment, rather than gross investment. Hence, adjustment costs are high when investment is higher, as well as lower than depreciated capital.

[32] Unlike Zhang (2005) and many other papers in this literature, our model does not have operating leverage (fixed cost of production). Operating leverage helps those models to magnify the differences between low and high productivity firms since they disproportionately affect the low productivity firms. Like Tüzel (2010); Bazdresch, Belo, and Lin (2009); and Tüzel and Jones (Forthcoming), our model features labor as a factor of production. Marginal product of labor across firms is equalized as labor is free to move between them. This leads to vastly different optimal hiring and investment decisions between firms and generates significant heterogeneity without incorporating operating leverage.

[33] For simplicity, we assume symmetric adjustment costs, but asymmetric adjustment costs as in Zhang (2005) or Tüzel (2010) would also break this symmetry and strengthen our results.

We assume that the volatility of the pricing kernel is a decreasing function of aggregate productivity; hence, Sharpe ratios (and discount rates) are higher in bad times. Risk is defined as the covariation of the returns with the pricing kernel. Higher volatility of the pricing kernel implies higher covariation with the kernel, hence higher risk in bad times. Lower rates of investment of low productivity firms tend to occur during recessions when Sharpe ratios are high, making them especially risky. Therefore, low productivity firms, during economic downturns, have the highest expected returns. These firms are the primary drivers of negative cross sectional relations between expected returns and firm productivity. Higher rates of investment of high productivity firms, on the other hand, tend to occur during expansionary periods when Sharpe ratios are low, and therefore result in lower expected returns for these firms.

In order to demonstrate the effect of aggregate shocks on low and high TFP firms over the business cycle, Figure 2 plots the sensitivity of low and high TFP firms' returns to aggregate TFP shocks conditional on the state of the economy. Good (bad) times are defined as times when the last period's aggregate TFP is more than one standard deviation above (below) its mean. Low (high) TFP firms are firms in the lowest (highest) TFP decile based on last year's ranking. The top panel plots the aggregate shock - realized return (averaged over all low or high TFP firms every period) relationship in good times, the bottom panel plots the same in bad times. There is a positive relationship between aggregate shocks and realized returns for all firms and economic environments. However, the relationship is quite flat for all firms in good times, whereas the sensitivity of all firms, but especially the low TFP firms, is much higher in bad times. These are the times when low TFP firms are the riskiest. Bad aggregate shocks lead to very negative returns, whereas good shocks drive the high returns of low TFP firms in bad times.

Table XII presents the results on the cross section of firms. Similar to our empirical analysis in Sections 1.3 and 1.4.1, we form decile portfolios by sorting firms on the basis of past TFP level. The table shows average firm characteristics and expected returns of these portfolios as well as the spreads between the low and high portfolios. We also present confidence intervals around the model-implied values of the spreads in returns. The results indicate that the model is able to match the data presented in Tables III and IV reasonably well. The parameters of the firm level TFP process are taken from the empirical estimates, so it is not surprising that the average TFP of the simulated portfolios are matched almost exactly to the data. However, the

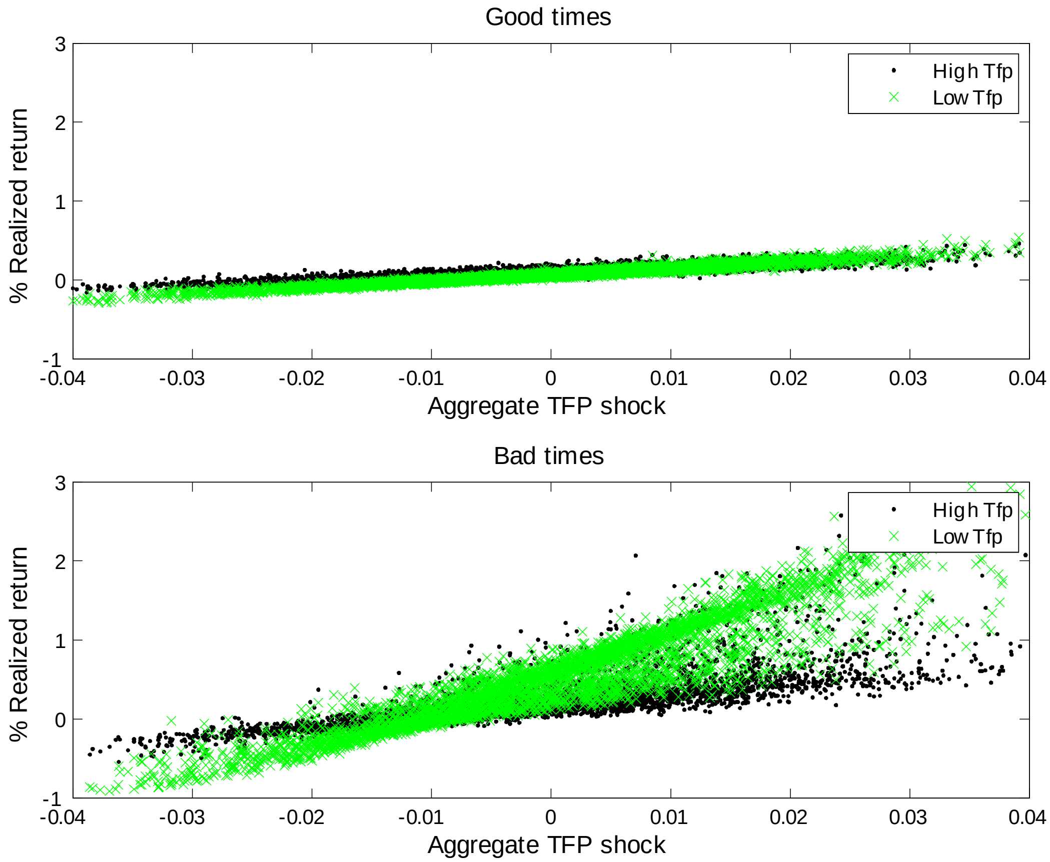

Figure 2: Sensitivity of firm returns to aggregate shocks. Top figure plots the realized returns as a function of aggregate productivity for low TFP and high TFP firms in good times. Bottom figure plots the realized returns in bad times.

model is not calibrated to match the cross section of the remaining characteristics, namely the investment to capital ratio, the hiring rate, firm size, book to market ratio, and the expected returns. We find that the investment to capital ratio and hiring rate both increase monotonically with TFP. Productive firms rationally invest and hire more than the unproductive firms since both capital and labor will be more efficient (their net present value would be higher) for those firms. The dispersion in investment to capital ratios generated by the model economy is reasonably close to what we observe in the data (0.3 versus 0.22), the dispersion in hiring rates of low versus high productivity firms is an order of magnitude bigger in the model.[34]

TFP is also monotonically and positively related to size and negatively related to B/M. In

[34]This problem could be alleviated by introducing adjustment costs in hiring/firing, which is currently assumed to be costless. Adjustment costs in hiring would reduce the volatility in hiring rate, and hence reduce the dispersion in the hiring rate of the most productive and least productive firms.

the model, firm size is the ex-dividend value of the firm, $V_{it} - D_{it}$, which is approximately equal to $q_t K_{t+1}$ (the amount of capital, K_{t+1}, $\times$ Tobin's q, q_{it}).[35] Tobin's q is linear in investment to capital ratio (Eq. 13), which is monotonically increasing in TFP. Likewise, the amount of capital, K_{t+1}, is increasing in TFP due to the positive relationship between investment and TFP and the persistence in productivity. Therefore, we expect to see a positive relationship between TFP and firm size. B/M, the ratio of book value to the market value of the firm, is measured as the amount of capital, K_{it+1}, divided by the ex-dividend value of the firm, $V_{it} - D_{it}$. Hence, B/M is approximately equal to the inverse of the Tobin's q (q_{it}) in the model.[36] This leads to a negative relationship between B/M and TFP. The model is quite successful in matching the magnitude of the dispersion in B/M (-0.63 versus -0.87). Also, the model is able to capture about half of the size dispersion found in the data (1.60 versus 3.53). These results indicate that TFP shock alone can account for a significant fraction of the cross sectional dispersion in firm characteristics found in the data.

Table XII also reports that the expected returns of TFP sorted portfolios decline monotonically with TFP. The spread in expected returns, -7.99%, is close to the empirical spread of -7.65% reported in Table IV. These returns are calculated by weighting the firms in the portfolios equally, but, value-weighting the portfolios leads to quantitatively very similar results. Furthermore, consistent with our empirical findings, both the average expected returns and the spread between the low and high TFP portfolio returns are much higher in contractions, compared to expansions. For this exercise, each year, we define periods where aggregate productivity is more than one standard deviation lower than the mean (based on short sample statistics) as contractions, and the remaining periods as expansions. This definition leads to designating roughly 15% of the sample period as contractions, which is in line with the frequency of contractionary periods in the data (96 out of 564 months in our sample period). The excess returns are measured in the following year. The model generates -6.74% and -11.08% spread in returns of TFP sorted portfolios in expansions and contractions, respectively. The empirical spreads in expansions and contractions, -5.46% and -18.28%, respectively, both fall within the wide confidence intervals around the model generated spread point estimates. However, as in Lin and Zhang (2012), the model does not generate a significant spread in unconditional

[35] In the presence of constant returns to scale, $V_{it} - D_{it} = q_t K_{t+1}$. With decreasing returns to scale, $V_{it} - D_{it}$ slightly exceeds $q_t K_{t+1}$.

[36] M/B would be exactly equal to Tobin's q when the production technology is constant returns to scale.

CAPM alphas, or Fama-French 3-factor alphas, hence we do not tabulate them.

Table XII: Model Implied Characteristics and Excess Returns for TFP Sorted Portfolios

	Low	2	3	4	5	6	7	8	9	High	High-Low Model	High-Low Data
TFP	0.52	0.68	0.78	0.87	0.95	1.05	1.16	1.29	1.48	1.93	1.41	1.35
I/K	0.01	0.02	0.03	0.04	0.06	0.07	0.10	0.12	0.17	0.30	0.30	0.22
HR	-0.25	0.00	0.17	0.33	0.49	0.67	0.88	1.14	1.53	2.60	2.84	0.25
Size	0.50	0.61	0.68	0.76	0.83	0.92	1.03	1.17	1.40	2.10	1.60	3.53
B/M	1.37	1.18	1.10	1.04	0.98	0.94	0.89	0.85	0.80	0.74	-0.63	-0.87
						Future Returns						
Mean Excess Returns												
	All states											
r^e	11.40	9.92	9.09	8.42	7.81	7.21	6.59	5.89	5.02	3.41	-7.99	-7.65
											(–12.88,-5.64)	
	Expansions											
r^e	6.82	5.54	4.83	4.25	3.73	3.22	2.69	2.11	1.39	0.08	-6.74	-5.46
											(-8.69,-5.52)	
	Contractions											
r^e	25.45	23.72	22.65	21.78	20.97	20.16	19.32	18.35	17.12	14.37	-11.08	-18.28
											(-21.53,-5.62)	

Note: The table reports the model-implied average characteristics (TFP, investment/capital ratio, hiring rate, firm size, book to market ratio), future mean excess returns of the TFP sorted portfolios. All values are based on 500 simulations of 1,000 firms for 50 periods (years). Point estimates are simulation averages, while confidence intervals (in parentheses) for the spreads in returns and alphas are constructed from the 5th and 95th percentiles of the simulated distributions of those spreads. Expansion and contraction periods are designated in year t based on the level of aggregate TFP in that year. Mean excess returns are measured in the following year. We define periods where aggregate productivity is more than one standard deviation lower than the mean (based on short sample statistics) as contractions, and the remaining periods as expansions. The equivalent empirical observations of High-Low portfolio from Tables III, IV, and V are given in the last column. All values are annual and in percentage terms.

2.5 Comparative Statics and Properties

Table XIII contains comparative statics for most parameters of our model. To better isolate the effect of each parameter while still roughly matching the time series targets, we focus on small changes in the parameter values. As is standard in models with convex adjustment costs, the adjustment cost parameter η primarily affects the volatility and risk of investment. Lower η

leads to higher investment volatility, lower risk, and lower expected returns in the economy. In the cross section, however, differences in investment behavior fueled by lower η leads to bigger dispersion between the low and high TFP firms.

Changing the pricing kernel parameters γ_0 and γ_1 yields predictable results. While the two parameters play different roles, an increase in the absolute value of either parameter will raise the equity premium, and return spreads rise in an approximately proportional manner. As a result of greater variation in expected returns, investment becomes slightly more volatile as well.

Finally, lowering the sensitivity of wages to aggregate productivity (lower ω) increases the risk and risk premia. This is due to the fact that wages do not fall as far in recessions or rise as much in expansions, causing firms to make bigger adjustments to the quantity of labor and leading to larger fluctuations in output. This effect also causes higher volatility in investment.

Table XIII: Comparative Statics

Case		$E(r^e)$	$Std\left(\frac{I}{K}\right)$	Decile portfolio spreads in $\frac{I}{K}$	HR	Size	B/M	$E(r^e)$
Benchmark		6.16	15.66	0.30	2.84	1.60	-0.63	-7.99
low η	$\eta = 4$	5.90	16.65	0.31	2.85	1.63	-0.64	-8.08
high η	$\eta = 5$	6.43	14.61	0.28	2.83	1.56	-0.63	-7.91
low γ_0	$\gamma_0 = 3.24$	6.01	15.23	0.29	2.84	1.57	-0.63	-7.23
high γ_0	$\gamma_0 = 3.30$	6.17	16.11	0.30	2.85	1.69	-0.64	-8.79
low γ_1	$\gamma_1 = -13.2$	6.07	15.60	0.29	2.84	1.59	-0.63	-7.86
low γ_1	$\gamma_1 = -13.4$	6.22	15.71	0.30	2.84	1.60	-0.64	-8.07
low ω	$\omega = 0.1$	6.19	15.80	0.30	2.84	1.66	-0.63	-7.97
high ω	$\omega = 0.3$	5.87	15.53	0.30	2.84	1.65	-0.63	-8.01

Note: The table presents the comparative statics for our benchmark calibration. The top row shows moments from the calibrated model, whose parameters are reported in Table XI. The rows beneath show the moments that result from changing one parameter to the value given. Moments include $E(r^e)$, the market risk premium, $Std\left(\frac{I}{K}\right)$, the average firm-level volatility of investment, and the average spreads in investment to capital ratio, hiring rate, size, B/M ratio, and mean excess returns on TFP-sorted portfolios. All results are obtained from 500 simulations of 1,000 firms for 50 periods (years).

Model parameters, calibrated as described in Section 2.3, often have implications for other statistics as well. The parameter γ_1 governs the degree of countercyclicality of the market price of risk and also has implications on the degree of predictability in stock returns. In order to test whether our benchmark value of -13.32 implies a realistic amount of predictability, we run return predictability regressions where annual excess stock returns are predicted by our state variable, aggregate TFP. We find that aggregate TFP predicts stock returns with 9.4% R^2 at annual horizon[37], which is in line with the stock predictability regression R^2's reported in the literature (Jones and Tüzel (2013)).

The adjustment cost parameter η also has additional implications for the economy. Lower adjustment costs would imply higher elasticity of investment rate to Tobin's q. Our benchmark value of 4.45 generates an investment - q elasticity of 2.8 around the steady state. Due to the poor empirical fit of the neoclassical investment equation, it is hard to find good empirical estimates for this elasticity. The results in Cummins, Hassett, and Oliner (2006) imply an elasticity around 1.2. Hence, our elasticity of 2.8 indicates a relatively low adjustment cost. With this calibration, the fraction of output that is lost due to capital adjustment costs is around 5%.

The functional form we assume for the equilibrium wage rate has implications for the cyclicality of aggregate employment in the economy. To compare the cyclical behavior generated by our model to the data, we compute the correlation of total employment growth with output growth, and the volatility of employment growth relative to output. We find that the moments generated by the model measure up to the data well. In the postwar U.S. data, this correlation is 0.9, and the relative volatility of employment growth is 1.1, while both of these numbers are around 1 using the simulated data.

3 Conclusion

This paper provides new evidence about the link between TFP and stock returns, offering an explanation to how firm characteristics can rationally predict returns. We estimate firm level TFP and show that it is strongly related to several firm characteristics such as size, the book to market ratio, investment, and hiring rate. TFP is negatively related to future stock returns, as well as ex-ante discount rates. A production based asset pricing model calibrated to match the estimated TFP distribution can replicate the empirical relationship between TFP, firm

[37] In this regression and other model results, we compute statistics over short samples (50 periods) and report the average value from 500 simulations.

characteristics, and stock returns reasonably well. Firms that receive repeated bad shocks (low TFP firms) end up being small firms and are characterized by low investment and hiring rates, and high B/M ratio. High TFP firms are large firms with high investment and hiring rates, and low B/M ratio. Low TFP firms end up being riskier than high TFP firms. Thus, differences in TFP generates the observed relationships between characteristics and stock returns.

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4 Appendix

4.1 Measuring TFP

The main contributions to measuring firm level TFP are by Olley and Pakes (1996) and Levinsohn and Petrin (2003). The key difference between the two methods is that Olley and Pakes (1996) use investment whereas Levinsohn and Petrin (2003) use materials used in production as a proxy for TFP. Since data on investment is readily available and often non-zero at the firm level but data on materials is not, we follow Olley and Pakes (1996) to estimate firm level productivities.

In our benchmark case, we estimate the production function based on labor and physical capital as inputs. The production technology is given by $y_{it} = F(l_{it}, k_{it}, \omega_{it})$ where y_{it} is log output for firm i in period t. l_{it}, k_{it} are log values of labor and capital of the firm. ω_{it} is the productivity, and η_{it} is an error term not known by the firm or the econometrician. Specifically:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \eta_{it}. \tag{16}$$

Olley and Pakes assume that productivity, ω_{it}, is observed by the firm before the firm makes some of its factor input decisions, which gives rise to the simultaneity problem. Labor, l_{it}, is the only variable input, i.e., its value can be affected by current productivity, ω_{it}. The other input, k_{it}, is a fixed input at time t, and its value is only affected by the conditional distribution of ω_{it} at time $t-1$. Consequently, ω_{it} is a state variable that affects firms' decision making where firms that observe a positive productivity shock in period t will invest more in capital, i_{it}, and hire more labor, l_{it}, in that period. The solution to the firm's optimization problem results in the equations for i_{it} :

$$i_{it} = i\left(\omega_{it}, k_{it}\right) \tag{17}$$

where both i and j are strictly increasing in ω. The inversion of the equations yield:

$$\omega_{it} = h(i_{it}, k_{it})$$

where h is strictly increasing in i_{it}.

Define:

$$\phi_{it} = \beta_0 + \beta_k k_{it} + h(i_{it}, k_{it}). \tag{18}$$

Using equations (16) and (18), we can obtain

$$y_{it} = \beta_l l_{it} + \phi_{it} + \eta_{it} \tag{19}$$

where we approximate ϕ_{it} with a second order polynomial series in capital and investment.[38] This first stage estimation results in an estimate for $\widehat{\beta}_l$ that controls for the simultaneity problem. In the second stage, consider the expectation of $y_{i,t+1} - \widehat{\beta}_l l_{i,t+1}$ on information at time t and survival of the firm:

$$\begin{aligned} E_t\left(y_{i,t+1} - \widehat{\beta}_l l_{i,t+1}\right) &= \beta_o + \beta_k k_{i,t+1} + E_t(\omega_{it+1}|\omega_{it}, survival) \\ &= \beta_o + \beta_k k_{i,t+1} + g(\omega_{it,}\widehat{P}_{survival,t}) \end{aligned} \tag{20}$$

where $\widehat{P}_{survival,t}$ denotes the probability of firm survival from time t to time $t+1$. The survival probability is estimated via a probit of a survival indicator variable on a polynomial expression containing capital and investment. We fit the following equation by nonlinear least squares:

$$y_{i,t+1} - \widehat{\beta}_l l_{i,t+1} = \beta_k k_{i,t+1} + \rho\omega_{it} + \tau\widehat{P}_{survival,t} + \eta_{i,t+1} \tag{21}$$

where ω_{it} is given by $\omega_{it} = \phi_{it} - \beta_0 - \beta_k k_{it}$ and is assumed to follow an AR(1) process.[39] At the end of this stage, $\widehat{\beta}_l$ and $\widehat{\beta}_k$ are estimated.

Finally, productivity is measured by:

$$P_{it} = \exp(y_{it} - \widehat{\beta}_o - \widehat{\beta}_l l_{it} - \widehat{\beta}_k k_{it}). \tag{22}$$

Since our data set covers different industries with different market structures and factor prices, we estimate equation (19) with industry specific time dummies and then subtract them from the left hand side of equation (20). Hence, our firm level TFPs are free of the effect of industry or aggregate TFP in any given year.[40] Our use of industry specific time dummies and

[38] Approximating with a higher order polynomial instead does not significantly change the results.

[39] Estimating an AR(2) process has almost no impact on the estimated labor and capital shares $(\widehat{\beta}_l, \widehat{\beta}_k)$.

[40] In another specification, we compute firm level TFPs without using industry dummies in our first stage

price deflators for investment remove the impact of possible embodied technological progress as modeled in Greenwood, Hercowitz, and Krusell (1997 and 2000) and Fernández-Villaverde and Rubio-Ramírez (2007).[41]

4.2 Data

The main data source for firm level productivity estimation is Compustat. We use the Compustat fundamental annual data from 1962 to 2009. We delete observations of financial firms (SIC classification between 6000 and 6999) and regulated firms (SIC classification between 4900 and 4999).[42] Our sample for production function estimation is comprised of all remaining firms in Compustat that have positive data on sales, total assets, number of employees, gross property, plant, and equipment, depreciation, accumulated depreciation, and capital expenditures. The sample period starts in 1962. The sample is an unbalanced panel with approximately 12,750 distinct firms; the total number of firm-year observations is approximately 128,000.[43]

The key variables for estimating the firm level productivity in our benchmark case are the firm level value added, employment, and physical capital.[44] Firm level data is supplemented with price index for Gross Domestic Product as deflator for the value added and price index for private fixed investment as deflator for investment and capital, both from the Bureau of Economic Analysis, and national average wage index from the Social Security Administration.

Value added (y_{it}) is computed as Sales - Materials, deflated by the GDP price deflator.[45]

estimation. We analyze the industry adjusted TFPs of firms, which are the log TFPs in excess of their industry averages. The stylized facts generated from that framework are both qualitatively and quantitatively very similar to our results even though the production function estimates for labor and capital are somewhat different.

[41] Examining cross sectional implications of differences in exposure to technological innovations as in Garleanu, Panageas, and Yu (2012), or implications of investment specific technological change by introducing vintage capital as in Kogan and Papanikolaou (2011) are left for future research.

[42] We exclude financial firms and regulated firms since it is standard to do so in this literature. Keeping these firms in the sample does not change the results in any material way. The production function estimates for the financial firms are quite similar to the production function estimates for the non-financial firms.

[43] At this stage, we do not require the firms to be in the CRSP database. Hence, our sample size gets somewhat smaller later when we merge our dataset with CRSP data.

[44] Firms use many inputs in their production, such as raw materials, labor, different types of capital, energy, etc. In our specification, we focus on labor and physical capital as the main inputs. Consequently, firm's value added is defined as the gross output net of expenditures on materials as well as the other expensed items such as advertisement, R&D expenditures, and rental expenses. Hence, our value added variable contains the contribution of labor and owned physical capital of the firm only.

[45] Measures of productivity based on firm revenues typically confound idiosyncratic demand and factor price effects with differences in efficiencies. Foster, Haltiwagner, and Syverson (2008) show that demand factors can be important in understanding industry dynamics and reallocation. Measures of productivity that incorporate demand factors require data on producers' physical outputs as well as product prices, which are not available at the firm level. However, they also show that revenue based productivity measures, such as the one used in this

Sales is net sales from Compustat (SALE).[46] Materials is measured as Total expenses minus Labor expenses. Total expenses is approximated as [Sales - Operating Income Before Depreciation and Amortization (Compustat (OIBDP))]. Labor expenses is calculated by multiplying the number of employees from Compustat (EMP) by average wages from the Social Security Administration).[47] The stock of labor (l_{it}) is measured by the number of employees from Compustat (EMP). These steps lead to our value added definition that is proxied by Operating Income Before Depreciation and Amortization+labor expenses.

Capital stock (k_{it}) is given by gross property, plant, and equipment (PPEGT) from Compustat, deflated by the price deflator for investment following the methods of Hall (1990) and Brynjolfsson & Hitt (2003).[48] Since investment is made at various times in the past, we need to calculate the average age of capital at every year for each company and apply the appropriate deflator (assuming that investment is made all at once in year [current year - age]). Average age of capital stock is calculated by dividing accumulated depreciation (Gross PPE - Net PPE, from Compustat (DPACT)) by current depreciation, from Compustat (DP). Age is further smoothed by taking a 3-year moving average.[49] The resulting capital stock is lagged by one period to measure the available capital stock at the beginning of the period.[50]

The Longitudinal Research Database (LRD), a large panel data set of U.S. manufacturing plants developed by the U.S. Bureau of the Census, is another dataset that is widely used in TFP estimations. One major shortcoming of the LRD for our purposes is that it excludes data on headquarters, sales offices, R&D labs, and the other auxiliary units that service manufacturing establishments of the same company. Such data is available from the Auxiliary Establishment

study, are highly correlated with their physical productivity.

[46] Net sales are equal to gross sales minus cash discounts, returned sales, etc.

[47] Compustat also has a data item called staff expense (XLR), which is sparsely populated. Comparing our labor expense series with the staff expense data available at Compustat reveals that our approximation yields a relatively correct and unbiased estimate of labor expenses.

[48] Hulten (1990) discusses many complications related to the measurement of capital. The principal options are to look for a direct estimate of the capital stock, K, or to adjust book values for inflation, mergers, and accounting procedures; or to use the perpetual inventory method to construct the capital stock from data on investments. There are problems associated with either method, and most of the time, the choice between these methods is dictated by the availability of data. Our results are insensitive to the treatment of inventories as a part of the capital stock.

[49] If there are less than three years of history for the firm, the average is taken over the available years.

[50] We do not have detailed deflators and wages for individual industries in our current benchmark estimation using the general Compustat sample. For the sample of manufacturing firms, detailed deflators and wages at the 4-digit SIC code level are available from the NBER-CES Database. Even though it is arguably better to use industry level deflators, the downside of this approach would be limiting the sample to 5,700 manufacturing firms, and the time frame to 1959-2005.

In our estimation, we use industry specific time dummies, which lessens the potential problems with using broad deflators to a great extent.

Survey but only at 5 year intervals. Since our focus is on examining the link between annual firm level TFP and stock returns, missing a potentially important part of the firm activities is not desirable. Another shortcoming of the LRD data is that it is strictly limited to manufacturing establishments; hence, the non-manufacturing sector, which is getting more important over time, is not represented at the LRD. Consequently, we use the Compustat data for measuring firm level TFP.

Fixed investment to capital ratio is given by firm level real capital investment divided by the beginning of the period real capital stock. Investment to capital ratio for organizational capital is obtained similarly. Asset growth is the percent change in total assets (TA) from Compustat. Hiring rate at time t is the change in the stock of labor (EMP) from time $t-1$ to t. Inventory growth is the percent change in inventories (INVT) from Compustat. R&D/PPE is the research and development expenditures (XRD from Compustat) divided by gross property, plant, and equipment. Real estate ratio for each firm is calculated by dividing the real estate components of PPE (sum of buildings and capitalized leases) by total PPE. Firm size is the market value of the firm's common equity (number of shares outstanding times share price from Center for Research in Security Prices (CRSP)). B/M, net stock issues (NS), and ROE are defined as in Fama and French (2008). Gross profitability is gross profits/total book assets as defined in Novy-Marx (Forthcoming). ROA is net income (income before extraordinary (Compustat item IB), minus dividends on preferred (item DVP), if available, plus income statement deferred taxes (item TXDI), if available) divided by total assets (item AT). Leverage is calculated by dividing long-term debt holdings (item DLTT in Compustat) by firm's total assets calculated as the sum of their long-term debt and the market value of their equity. Firm age (AGE) is proxied by the number of years since the firm's first year of observation in Compustat.

4.3 Robustness Checks

We examine the sensitivity of our production function estimates, our measure of firm level TFPs, and the resulting relationship between firm level TFPs, firm characteristics, and returns to a large number of alternative specification. On the measurement of inputs, we experiment with defining the capital stock inclusive of inventories as in Cooley or Prescott (1995), broadening the Olley and Pakes method to include organization capital as another input to the production function[51], using a broad definition of fixed capital that includes the R&D capital, as well as using different deflators and prices (such as industry deflators), carrying out the estimation at the industry level (allowing for different production function estimates for industries). Our overall results on the relationship between TFP and firm characteristics and returns are not sensitive to any of these specifications. The results are also not sensitive to carrying out the estimation with industry specific time dummies at 2, 3, and 4 digit SIC levels. The findings are also similar for manufacturing versus non-manufacturing firms and over different sample periods.

Following Liu, Whited, and Zhang (2009), we also compute unlevered equity returns and study the relationship between firm TFPs and future unlevered returns. Both unlevered TFP sorted portfolio returns and Fama-Macbeth regressions with unlevered returns yield results that resemble the ones for levered returns. Hence, we confirm that differences in leverage are not the underlying reason for the spread between the low and high TFP equity returns found previously.

Like most variables that predict returns, our results based on realized returns are typically stronger for smaller firms. Following Fama and French (2008), which puts a special emphasis on micro cap firms, we replicated our full analysis by excluding micro cap firms (as defined by Fama and French, using the 20th percentile of the NYSE firm size distribution as the breakpoint) from our sample. As Fama and French (2008) also point out, this leads to eliminating more than half of the sample. In this smaller sample, we find that the magnitude of the spreads and coefficients are typically smaller (about half) and less significant (only at 10% level) in tests that use realized returns, though the results are still strong in contractions. But the results of tests that use implied cost of capital remain mostly unchanged.[52]

[51] We measure organizational capital based on data on firm's reported Sales, General, and Administrative expenses from Compustat (XSGA) as in Eisfeldt and Papanikolau (2009). We construct it by using the perpetual inventory method where XSGA is deflated by the price deflator for investment for the matching industry from the NBER-CES Database (PIINV) and assumed to depreciate by 20% per year.

[52] Detailed robustness results were presented in the earlier versions of the paper and are available from the authors upon request.